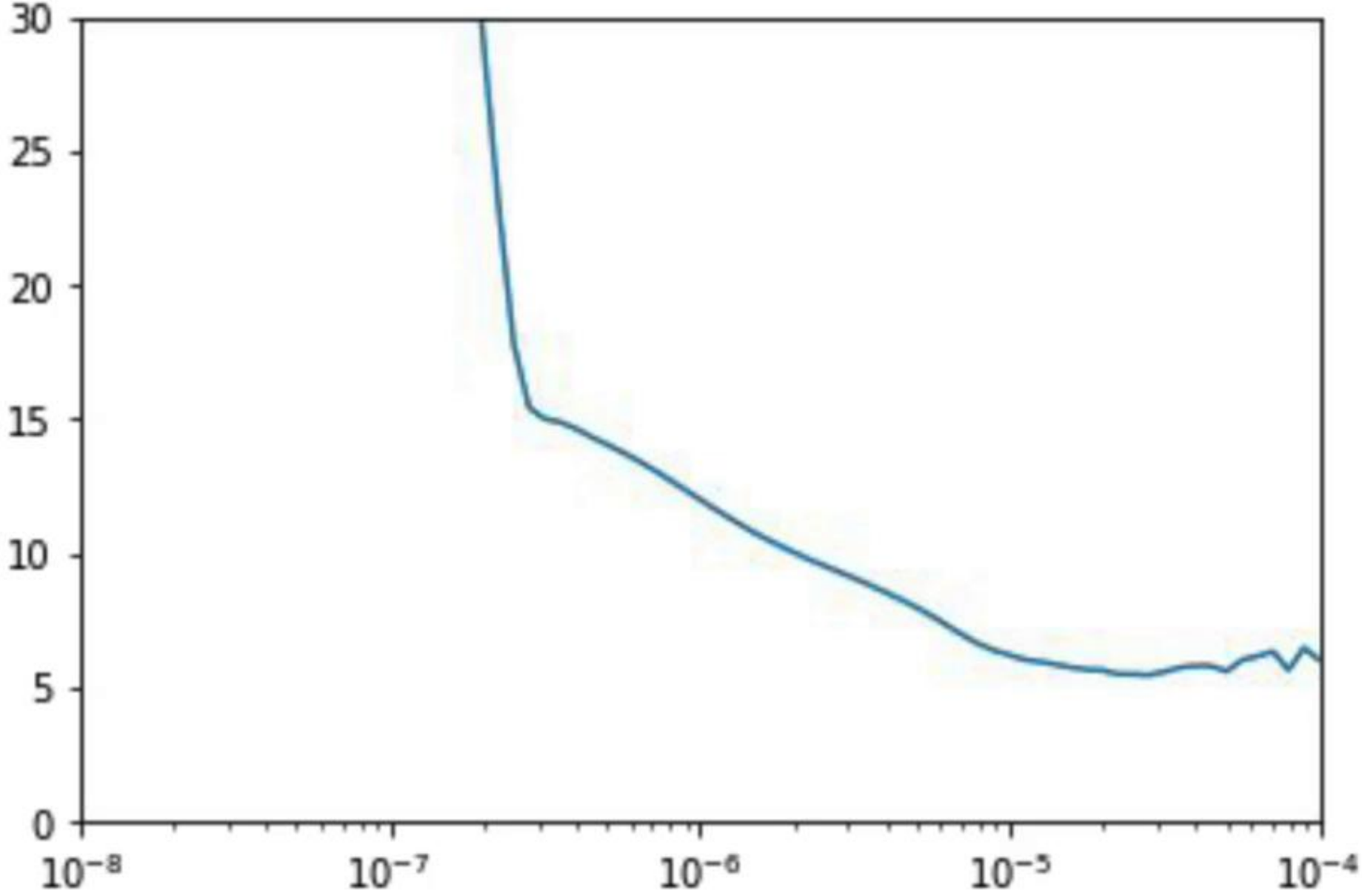
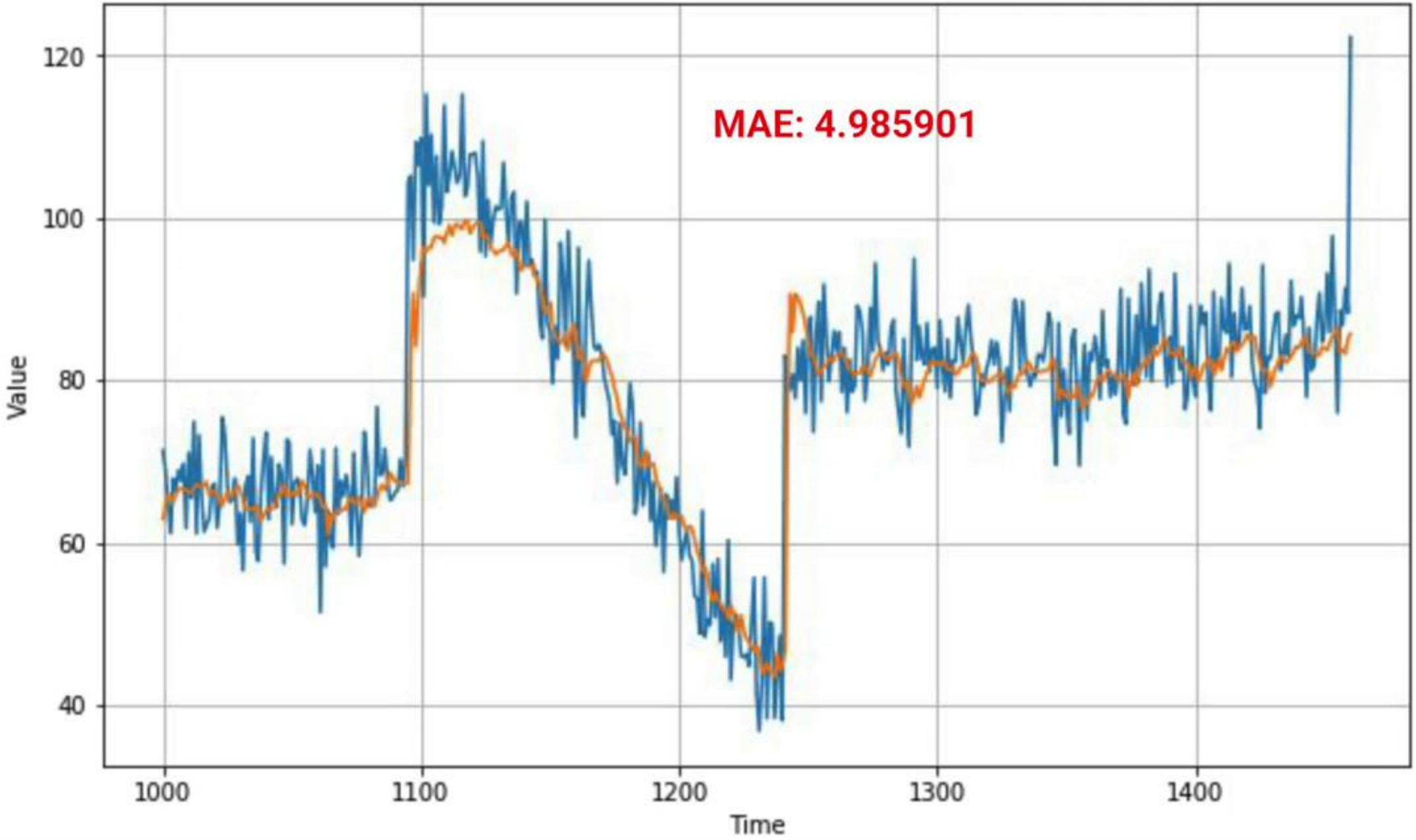
```python
model = tf.keras.models.Sequential([
  tf.keras.layers.Conv1D(filters=32, kernel_size=5,
                      strides=1, padding="causal",
                      activation="relu",
                      input_shape=[None, 1]),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.Dense(1),
  tf.keras.layers.Lambda(lambda x: x * 200)
])

optimizer = tf.keras.optimizers.SGD(lr=1e-5, momentum=0.9)

model.compile(loss=tf.keras.losses.Huber(),
              optimizer=optimizer,
              metrics=["mae"])

model.fit(dataset,epochs=500)
```

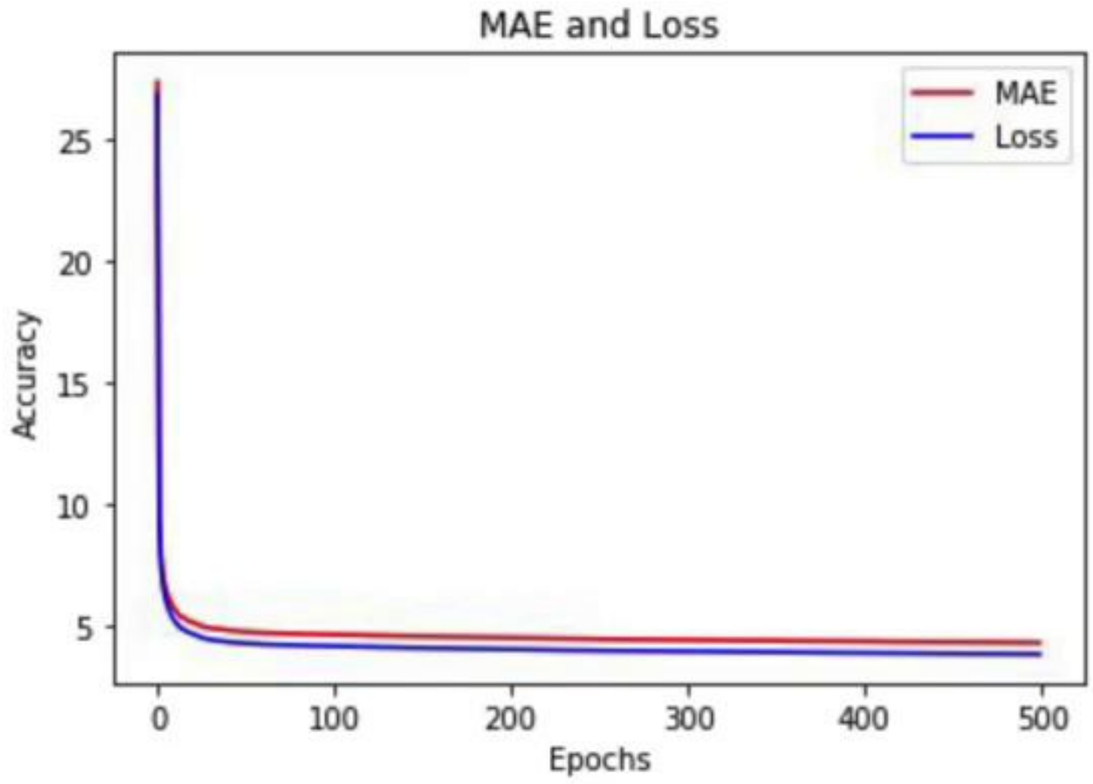
```
model = tf.keras.models.Sequential([
  tf.keras.layers.Conv1D(filters=32, kernel_size=5,
                         strides=1, padding="causal",
                         activation="relu",
                         input_shape=[None, 1]),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.Dense(1),
  tf.keras.layers.Lambda(lambda x: x * 200)
])

optimizer = tf.keras.optimizers.SGD(lr=1e-5, momentum=0.9)

model.compile(loss=tf.keras.losses.Huber(),
              optimizer=optimizer,
              metrics=["mae"])

model.fit(dataset,epochs=500)
```

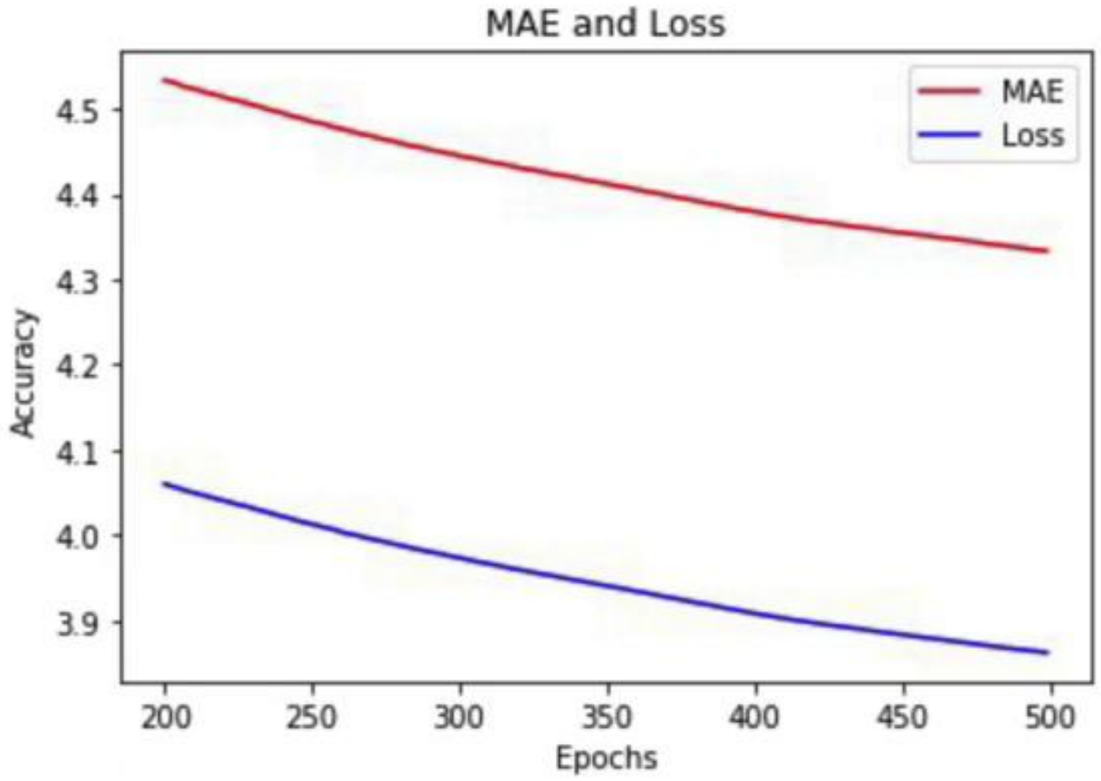
```python
def windowed_dataset(series, window_size, batch_size, shuffle_buffer):
    series = tf.expand_dims(series, axis=-1)

    ds = tf.data.Dataset.from_tensor_slices(series)
    ds = ds.window(window_size + 1, shift=1, drop_remainder=True)
    ds = ds.flat_map(lambda w: w.batch(window_size + 1))
    ds = ds.shuffle(shuffle_buffer)
    ds = ds.map(lambda w: (w[:-1], w[1:]))

    return ds.batch(batch_size).prefetch(1)
```


30
25
20
15
10
5
0
10^{-8}
10^{-7}
10^{-6}
10^{-5}
10^{-4}

```
model = tf.keras.models.Sequential([
  tf.keras.layers.Conv1D(filters=32, kernel_size=5,
                         strides=1, padding="causal",
                         activation="relu",
                         input_shape=[None, 1]),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.Dense(1),
  tf.keras.layers.Lambda(lambda x: x * 200)
])

optimizer = tf.keras.optimizers.SGD(lr=1e-5, momentum=0.9)

model.compile(loss=tf.keras.losses.Huber(),
              optimizer=optimizer,
              metrics=["mae"])

model.fit(dataset,epochs=500)
```

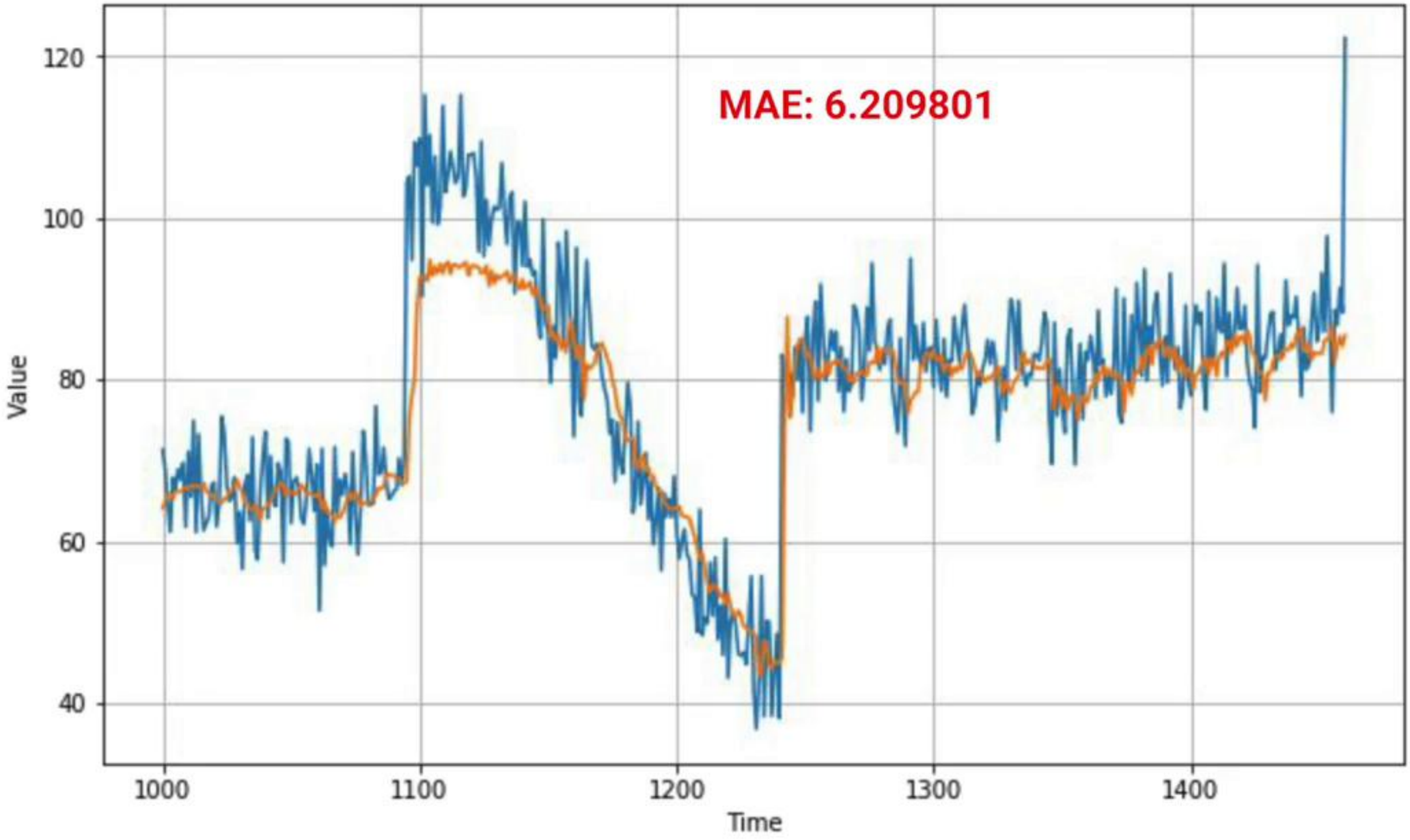

MAE: 4.985901
120
100
80
60
40
Value
1000
1100
1200
1300
1400
Time

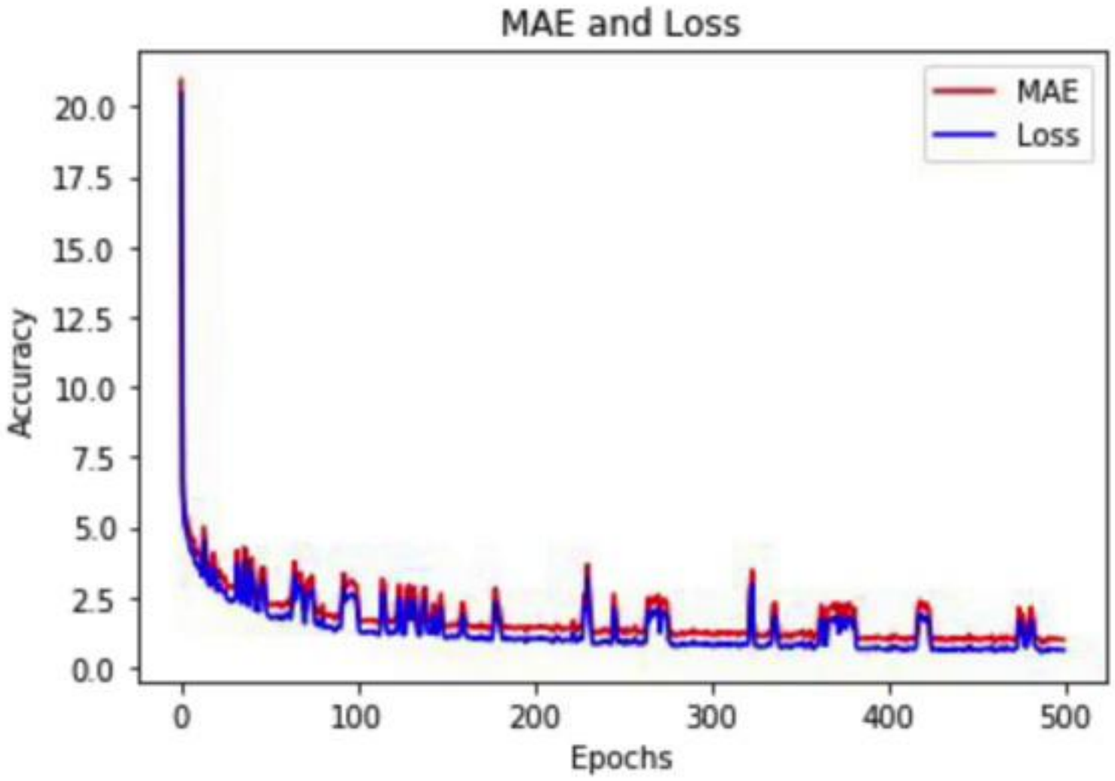

MAE and Loss
MAE
Loss
Accuracy
Epochs
25
20
15
10
5
0
100
200
300
400
500

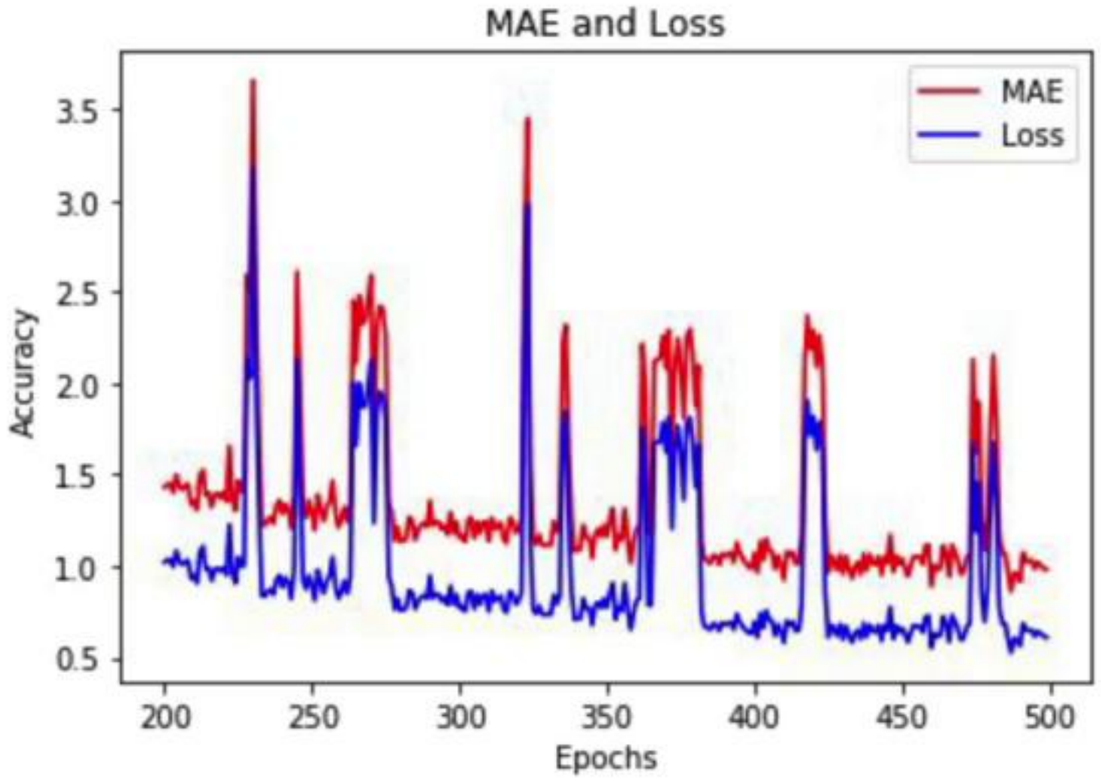

MAE and Loss
MAE
Loss
Accuracy
Epochs
4.5
4.4
4.3
4.2
4.1
4.0
3.9
200
250
300
350
400
450
500

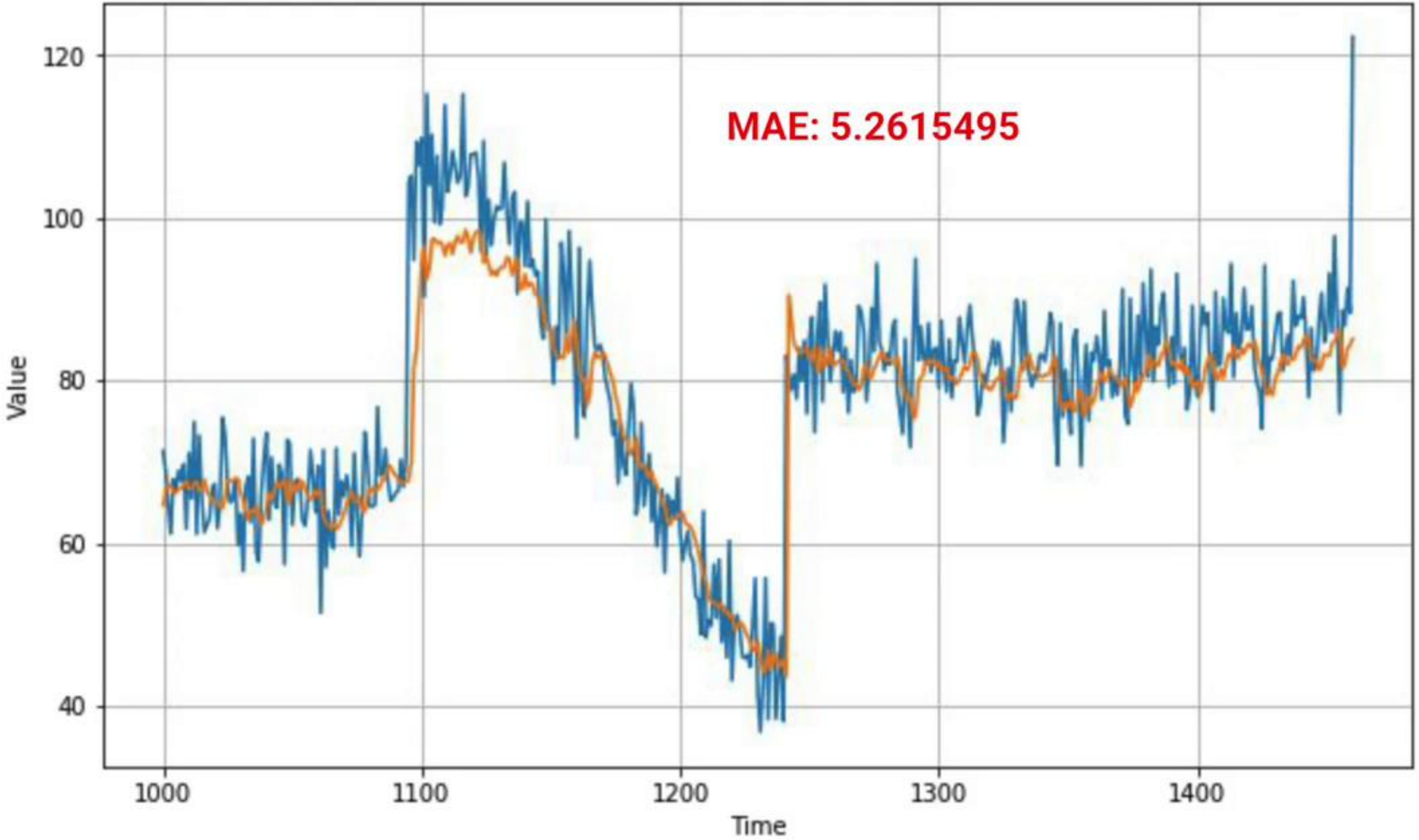
```
model = tf.keras.models.Sequential([
  tf.keras.layers.Conv1D(filters=32, kernel_size=5,
                         strides=1, padding="causal",
                         activation="relu",
                         input_shape=[None, 1]),
  tf.keras.layers.Bidirectional(tf.keras.layers.LSTM(32, return_sequences=True)),
  tf.keras.layers.Bidirectional(tf.keras.layers.LSTM(32, return_sequences=True)),
  tf.keras.layers.Dense(1),
  tf.keras.layers.Lambda(lambda x: x * 200)
])
```

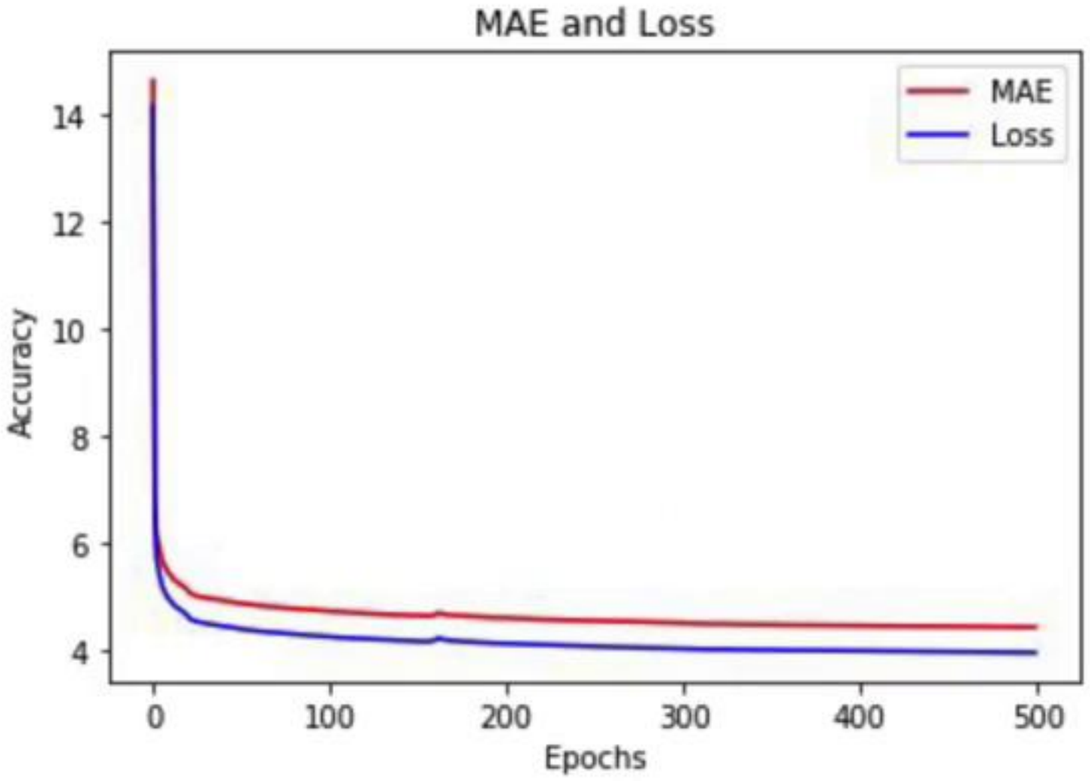
```
Epoch 495/500
31/31 [==============================] - 2s 58ms/step - loss: 0.6491 - mae: 1.0314
Epoch 496/500
31/31 [==============================] - 2s 60ms/step - loss: 0.6155 - mae: 0.9857
Epoch 497/500
31/31 [==============================] - 2s 59ms/step - loss: 0.6425 - mae: 1.0207
Epoch 498/500
31/31 [==============================] - 2s 59ms/step - loss: 0.6330 - mae: 1.0046
Epoch 499/500
31/31 [==============================] - 2s 59ms/step - loss: 0.6155 - mae: 0.9877
Epoch 500/500
31/31 [==============================] - 2s 59ms/step - loss: 0.6111 - mae: 0.9806
```

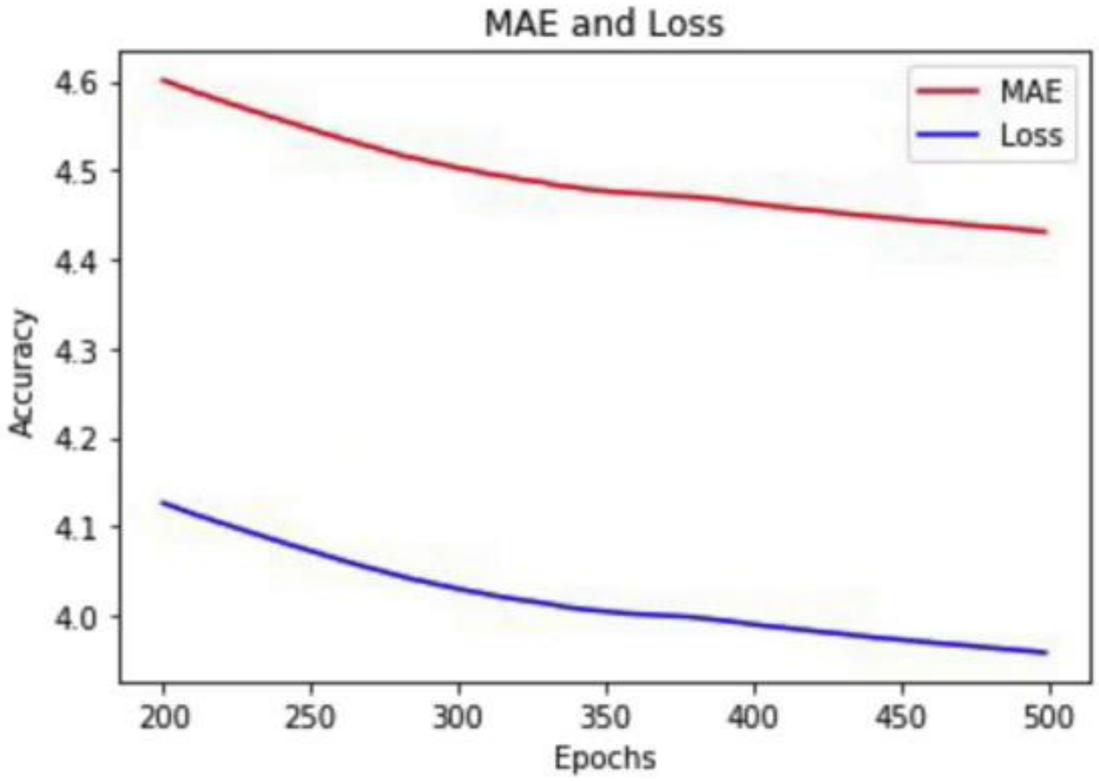

MAE: 6.209801
120
100
80
60
40
Value
1000
1100
1200
1300
1400
Time


MAE and Loss
MAE
Loss
Accuracy
Epochs
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0
100
200
300
400
500


MAE and Loss
MAE
Loss
Accuracy
Epochs
3.5
3.0
2.5
2.0
1.5
1.0
0.5
200
250
300
350
400
450
500


MAE: 5.2615495
120
100
80
60
40
Value
1000
1100
1200
1300
1400
Time


MAE and Loss
MAE
Loss
Accuracy
Epochs
14
12
10
8
6
4
0
100
200
300
400
500


MAE and Loss
MAE
Loss
Accuracy
Epochs
4.6
4.5
4.4
4.3
4.2
4.1
4.0
200
250
300
350
400
450
500

https://www.kaggle.com/robervalt/sunspots

Dataset

9

Sunspots

Monthly Mean Total Sunspot Number - form 1749 to july 2018

Especuloide • updated a year ago (Version 3)

Data | Kernels (5) | Discussion | Activity | Metadata | Download (22 KB) | New Kernel

Usability 7.1 | License CC0: Public Domain | Tags No tags yet

Description

Context

Sunspots are temporary phenomena on the Sun's photosphere that appear as spots darker than the surrounding areas. They are regions of reduced surface temperature caused by concentrations of magnetic field flux that inhibit convection. Sunspots usually appear in pairs of opposite magnetic polarity. Their number varies according to the approximately 11-year solar cycle.

Source: https://en.wikipedia.org/wiki/Sunspot

Content :

Data (22 KB)

Data Sources	About this file	Columns
Sunspots.csv 3235 x 3	Sunspots - Monthly Mean Total Sunspot Number	#
		Date
		Monthly Mean Total Sunspot Number

Sunspots.csv

```
,Date,Monthly Mean Total Sunspot Number
0,1749-01-31,96.7
1,1749-02-28,104.3
2,1749-03-31,116.7
3,1749-04-30,92.8
4,1749-05-31,141.7
5,1749-06-30,139.2
6,1749-07-31,158.0
7,1749-08-31,110.5
8,1749-09-30,126.5
9,1749-10-31,125.8
10,1749-11-30,264.3
11,1749-12-31,142.0
12,1750-01-31,122.2
13,1750-02-28,126.5
14,1750-03-31,148.7
15,1750-04-30,147.2
16,1750-05-31,150.0
17,1750-06-30,166.7
```

```
!wget --no-check-certificate \
    https://storage.googleapis.com/laurencemoroney-blog.appspot.com/Sunspots.csv \
    -O /tmp/sunspots.csv
```

```python
import csv
time_step = []
sunspots = []

with open('/tmp/sunspots.csv') as csvfile:
  reader = csv.reader(csvfile, delimiter=',')
  next(reader)
  for row in reader:
    sunspots.append(float(row[2]))
    time_step.append(int(row[0]))
```

```python
import csv
time_step = []
sunspots = []

with open('/tmp/sunspots.csv') as csvfile:
  reader = csv.reader(csvfile, delimiter=',')
  next(reader)
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Sunspots.csv

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0,1749-01-31,96.7
1,1749-02-28,104.3
2,1749-03-31,116.7
3,1749-04-30,92.8
4,1749-05-31,141.7
5,1749-06-30,139.2
6,1749-07-31,158.0
7,1749-08-31,110.5
8,1749-09-30,126.5
9,1749-10-31,125.8
10,1749-11-30,264.3
11,1749-12-31,142.0
12,1750-01-31,122.2
13,1750-02-28,126.5
14,1750-03-31,148.7
15,1750-04-30,147.2
16,1750-05-31,150.0
17,1750-06-30,166.7
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Sunspots.csv

```
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0,1749-01-31,96.7
1,1749-02-28,104.3
2,1749-03-31,116.7
3,1749-04-30,92.8
4,1749-05-31,141.7
5,1749-06-30,139.2
6,1749-07-31,158.0
7,1749-08-31,110.5
8,1749-09-30,126.5
9,1749-10-31,125.8
10,1749-11-30,264.3
11,1749-12-31,142.0
12,1750-01-31,122.2
13,1750-02-28,126.5
14,1750-03-31,148.7
15,1750-04-30,147.2
16,1750-05-31,150.0
17,1750-06-30,166.7
```

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  for row in reader:
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    time_step.append(int(row[0]))
```
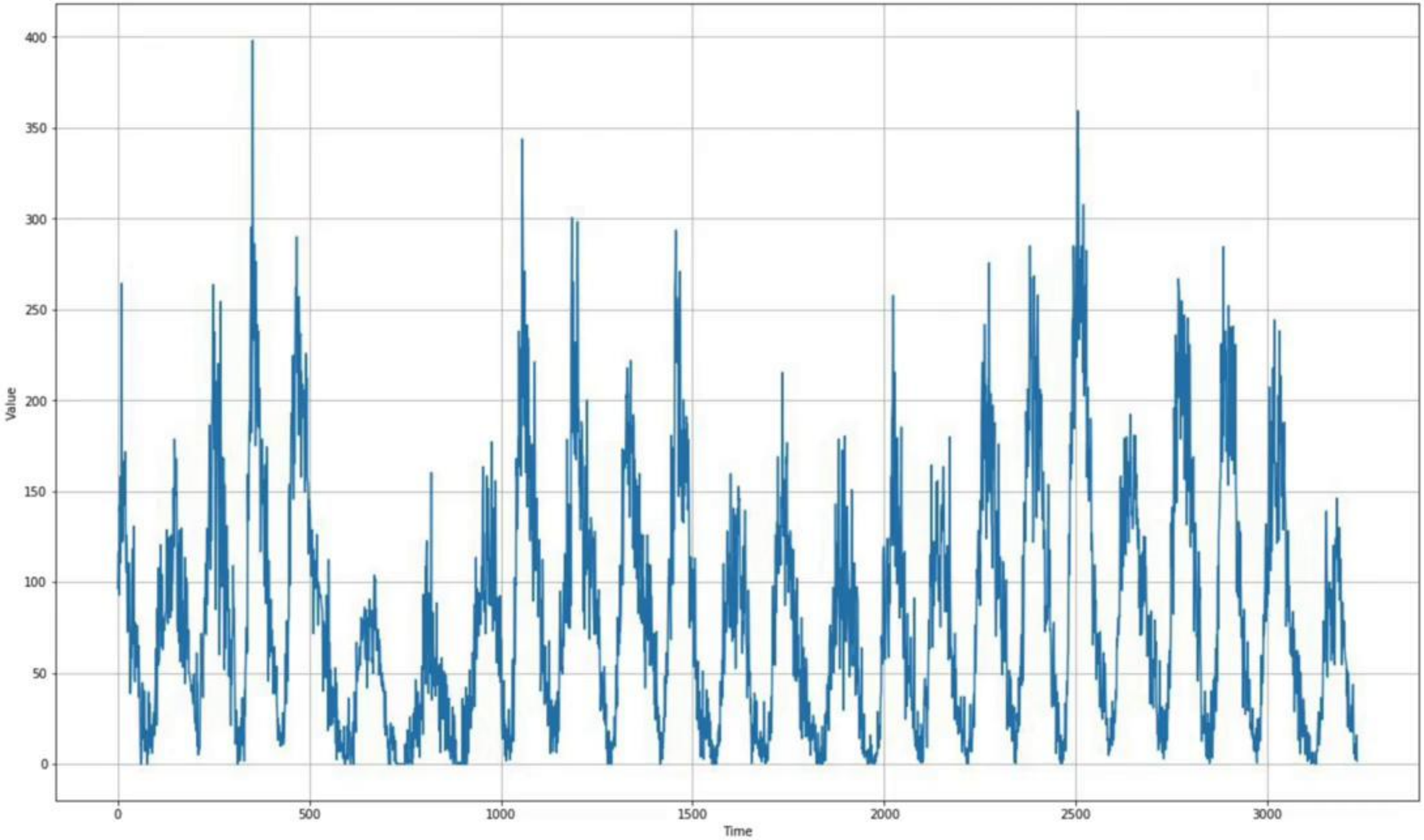

Sunspots.csv

```
,Date,Monthly Mean Total Sunspot Number
0,1749-01-31,96.7
1,1749-02-28,104.3
2,1749-03-31,116.7
3,1749-04-30,92.8
4,1749-05-31,141.7
5,1749-06-30,139.2
6,1749-07-31,158.0
7,1749-08-31,110.5
8,1749-09-30,126.5
9,1749-10-31,125.8
10,1749-11-30,264.3
11,1749-12-31,142.0
12,1750-01-31,122.2
13,1750-02-28,126.5
14,1750-03-31,148.7
15,1750-04-30,147.2
16,1750-05-31,150.0
17,1750-06-30,166.7
```

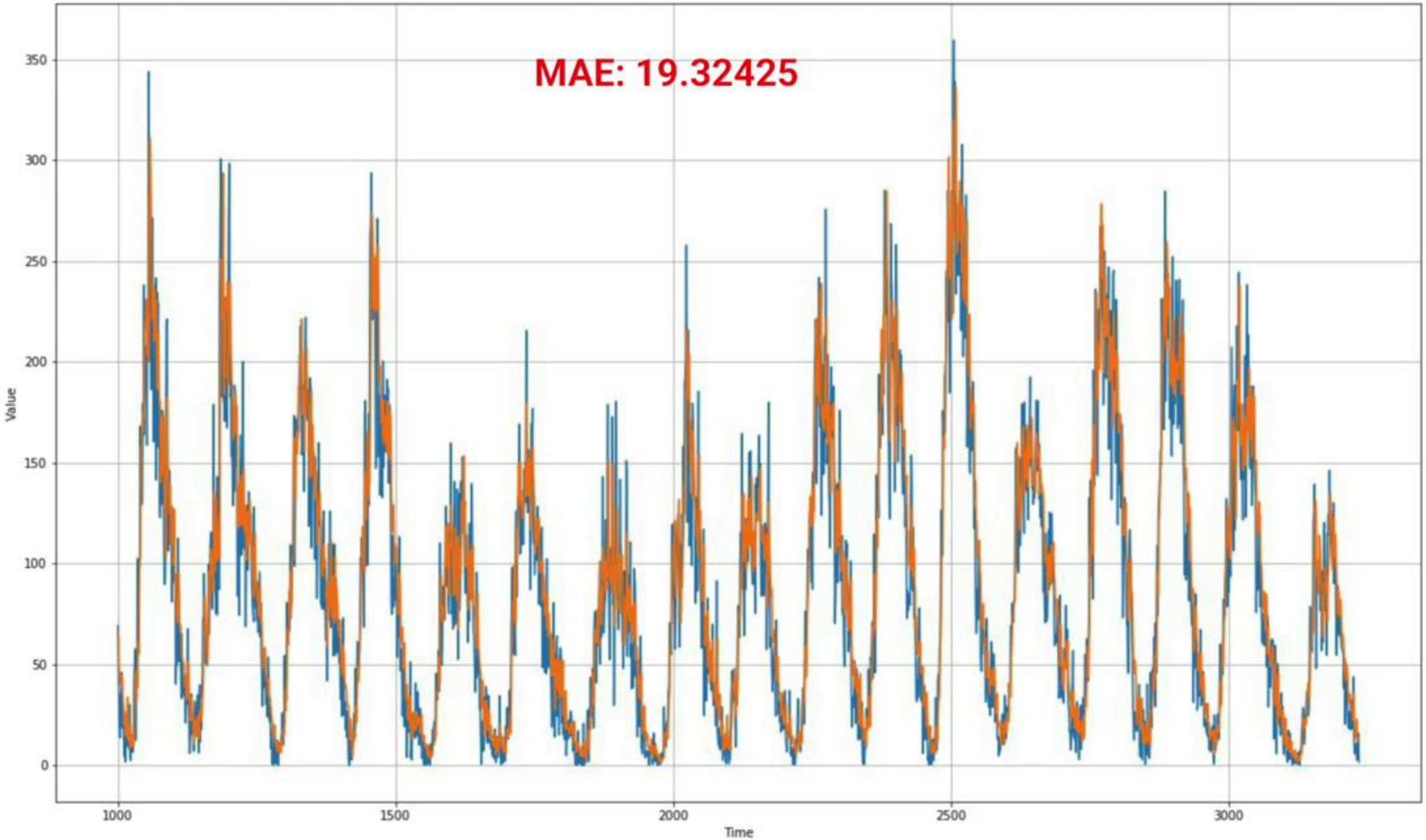
```
series = np.array(sunspots)
time = np.array(time_step)
```

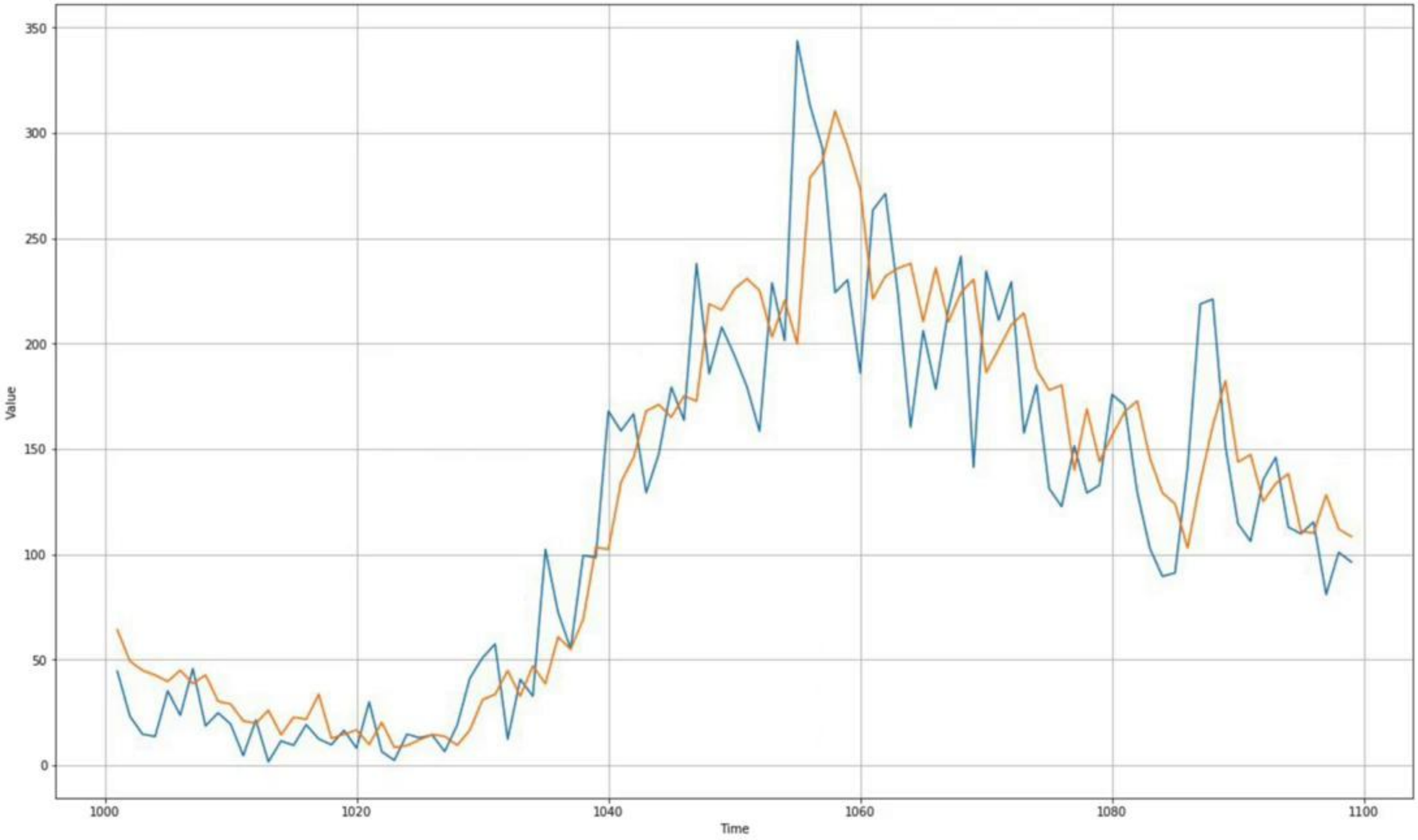

400
350
300
250
200
150
100
50
0
Value
0
500
1000
1500
2000
2500
3000
Time

```
split_time = 1000
time_train = time[:split_time]
x_train = series[:split_time]
time_valid = time[split_time:]
x_valid = series[split_time:]

window_size = 20
batch_size = 32
shuffle_buffer_size = 1000
```

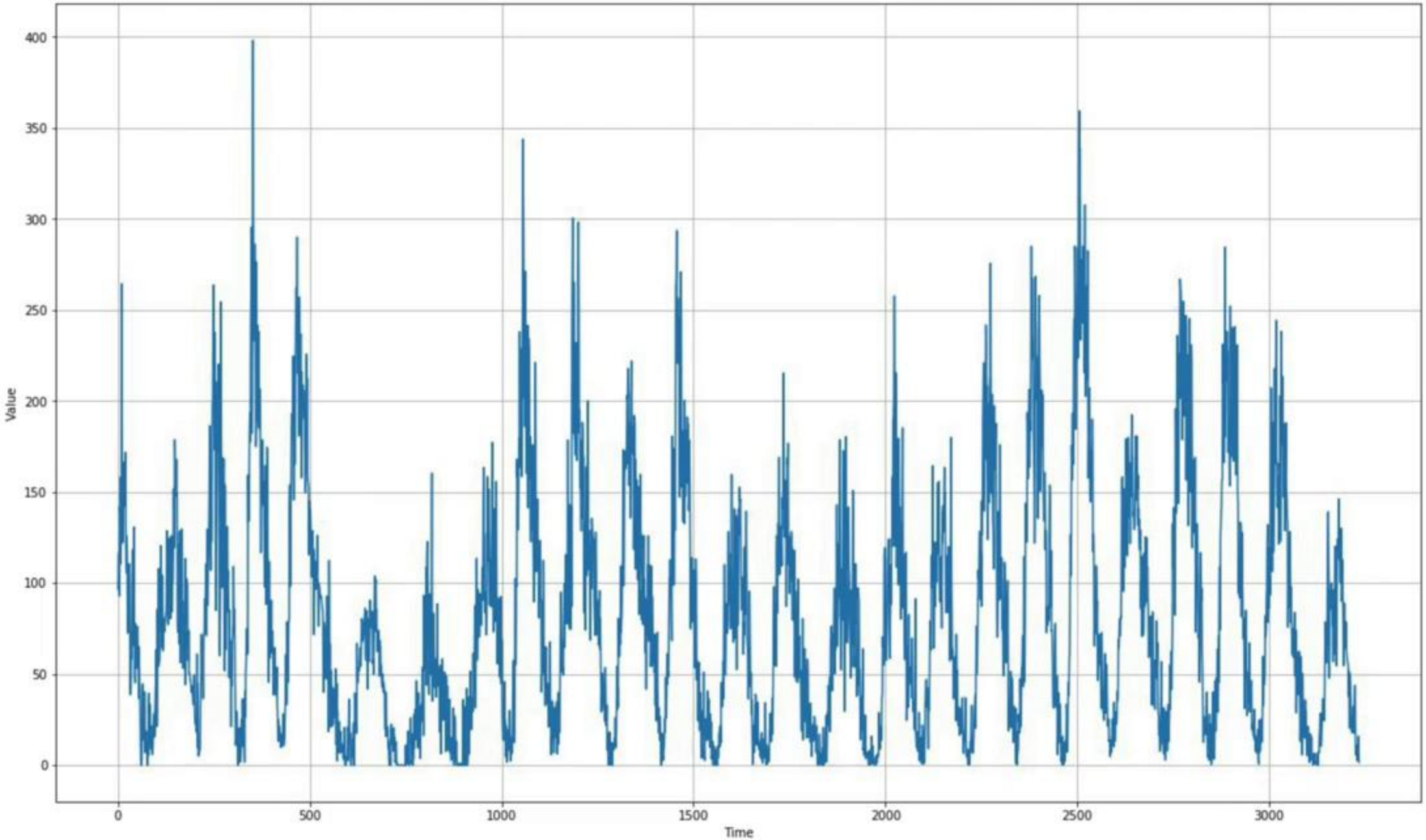
```python
def windowed_dataset(series, window_size, batch_size, shuffle_buffer):
  dataset = tf.data.Dataset.from_tensor_slices(series)
  dataset = dataset.window(window_size + 1, shift=1, drop_remainder=True)
  dataset = dataset.flat_map(lambda window: window.batch(window_size + 1))
  dataset = dataset.shuffle(shuffle_buffer)
                   .map(lambda window: (window[:-1], window[-1]))
  dataset = dataset.batch(batch_size).prefetch(1)
  return dataset
```

```
dataset = windowed_dataset(x_train, window_size, batch_size, shuffle_buffer_size)

model = tf.keras.models.Sequential([
    tf.keras.layers.Dense(10, input_shape=[window_size], activation="relu"),
    tf.keras.layers.Dense(10, activation="relu"),
    tf.keras.layers.Dense(1)
])

model.compile(loss="mse", optimizer=tf.keras.optimizers.SGD(lr=1e-6, momentum=0.9))
model.fit(dataset,epochs=100,verbose=0)
```

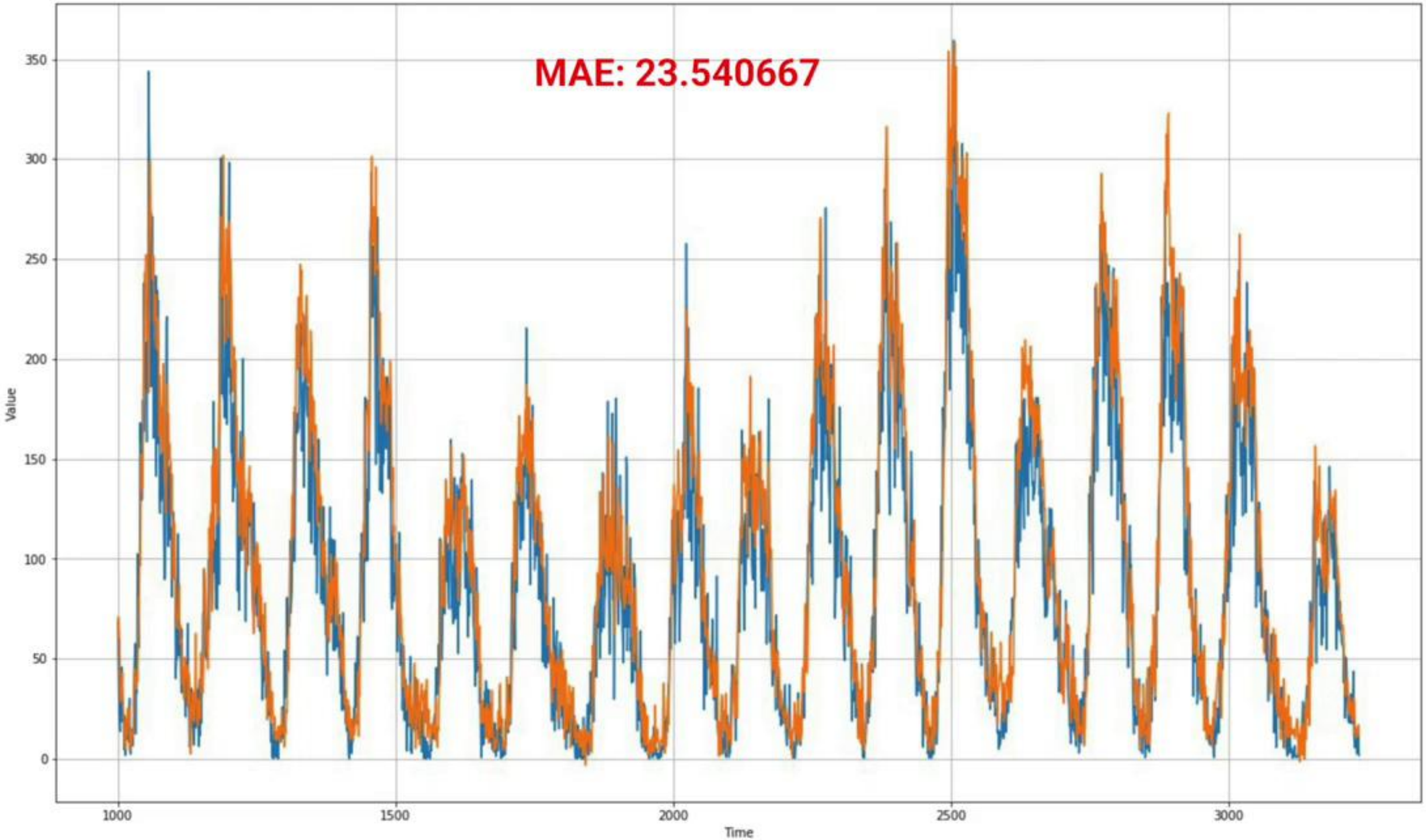

MAE: 19.32425
350
300
250
200
150
100
50
0
Value
1000
1500
2000
2500
3000
Time

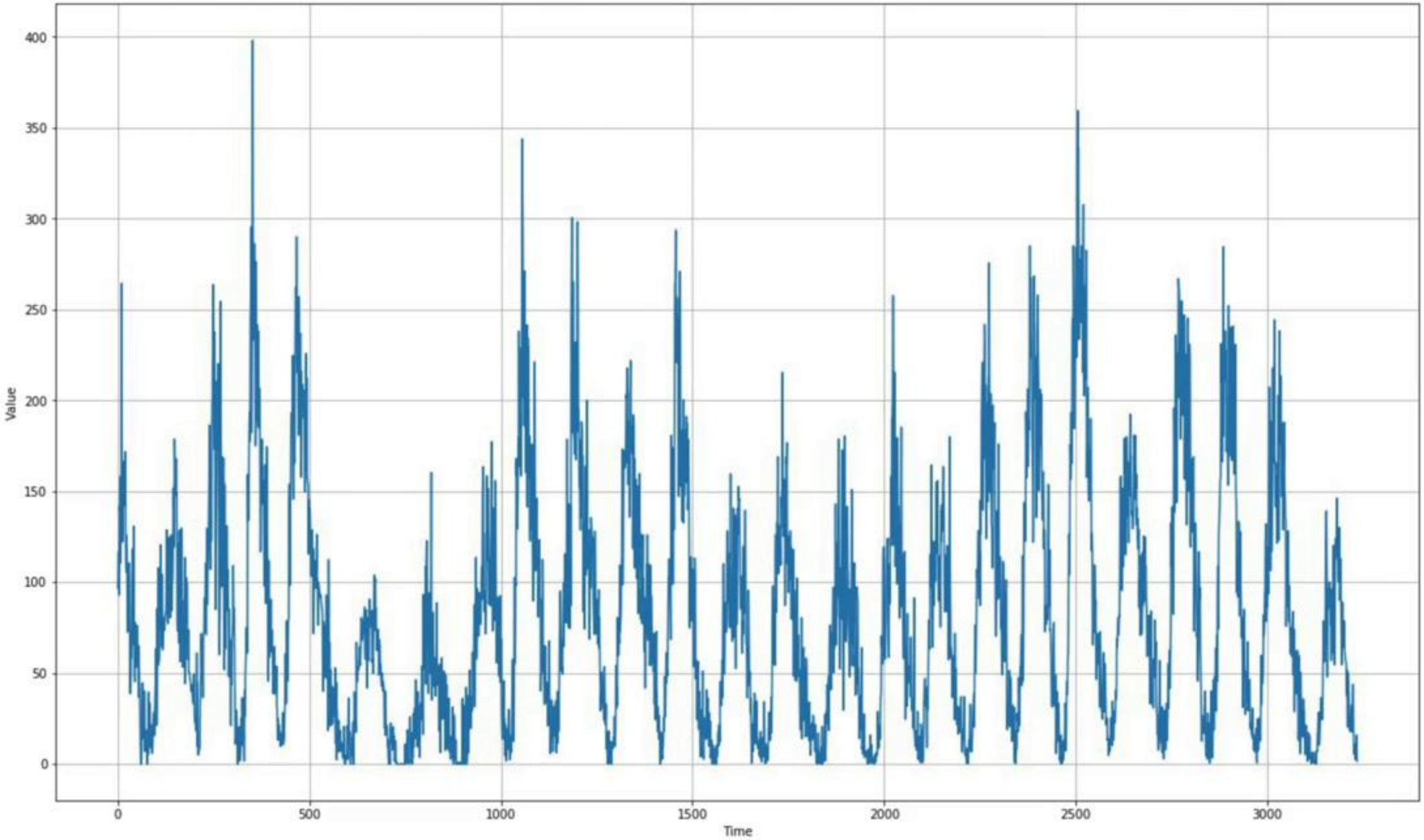

350
300
250
200
150
100
50
0
Value
1000
1020
1040
1060
1080
1100
Time

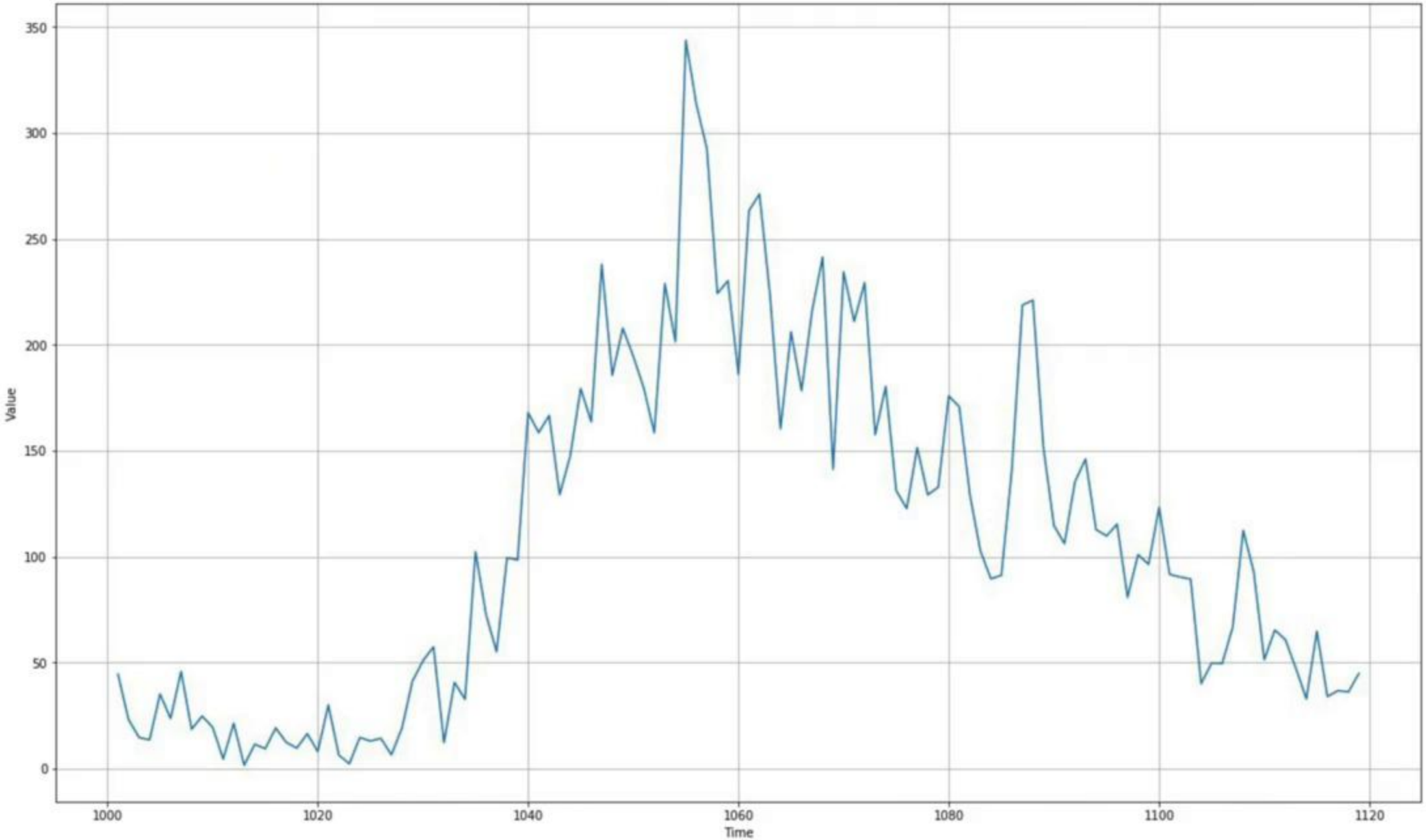
```
split_time = 1000
time_train = time[:split_time]
x_train = series[:split_time]
time_valid = time[split_time:]
x_valid = series[split_time:]

window_size = 20
batch_size = 32
shuffle_buffer_size = 1000
```


400
350
300
250
200
150
100
50
0
Value
0
500
1000
1500
2000
2500
3000
Time

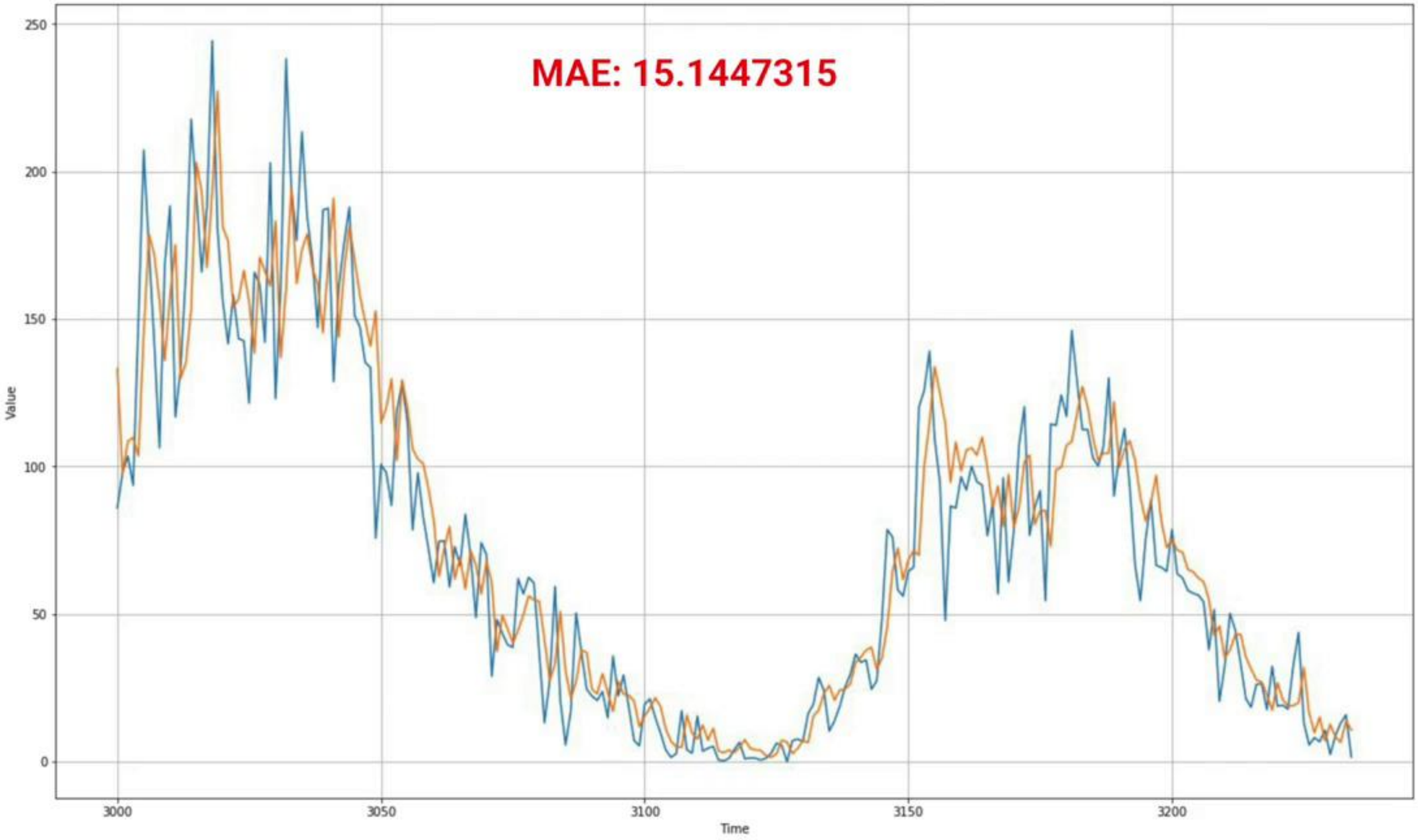
```
split_time = 1000
time_train = time[:split_time]
x_train = series[:split_time]
time_valid = time[split_time:]
x_valid = series[split_time:]

window_size = 132
batch_size = 32
shuffle_buffer_size = 1000
```


MAE: 23.540667
350
300
250
200
150
100
50
0
Value
1000
1500
2000
2500
3000
Time

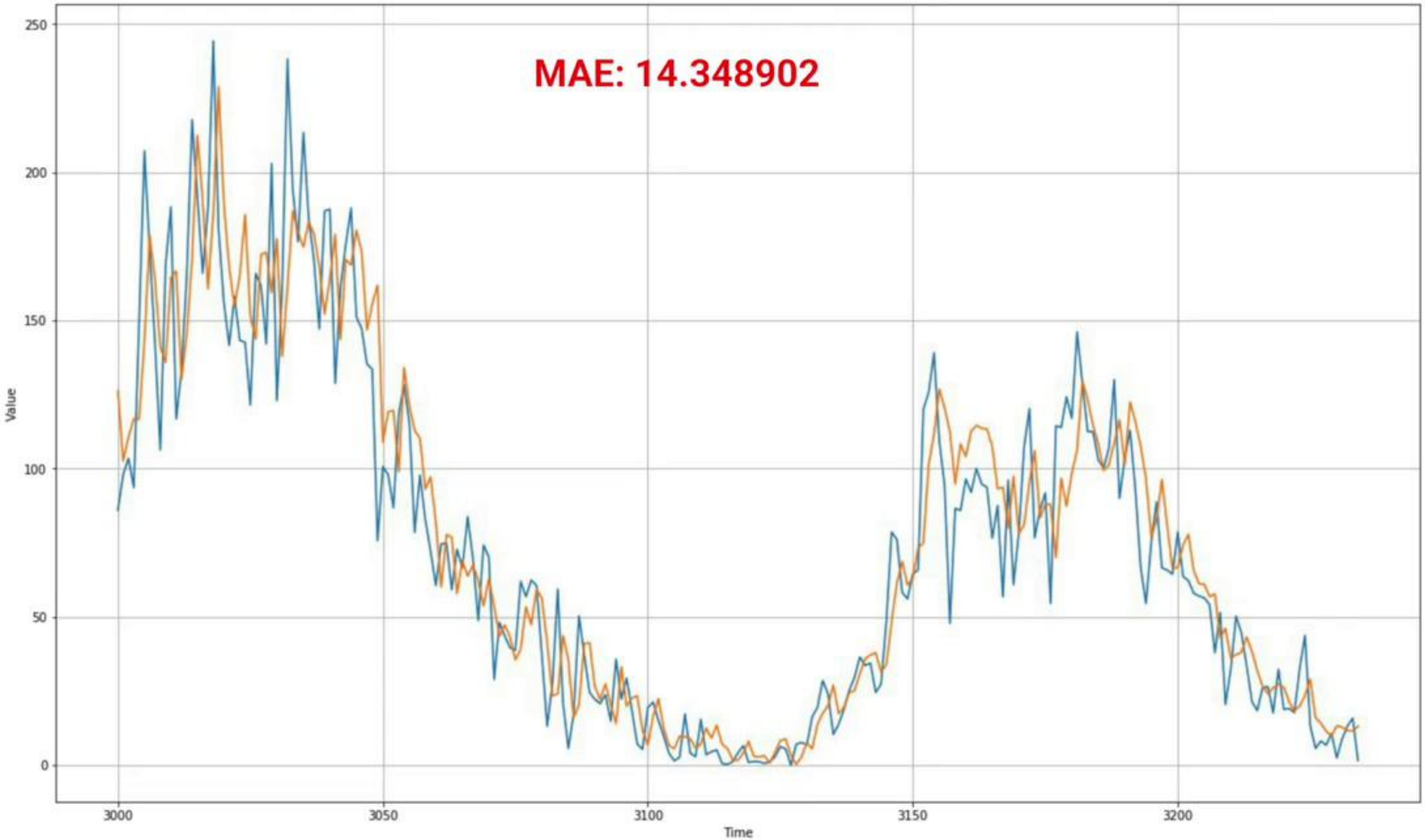

400
350
300
250
200
150
100
50
0
Value
0
500
1000
1500
2000
2500
3000
Time


350
300
250
200
150
100
50
0
Value
1000
1020
1040
1060
1080
1100
1120
Time

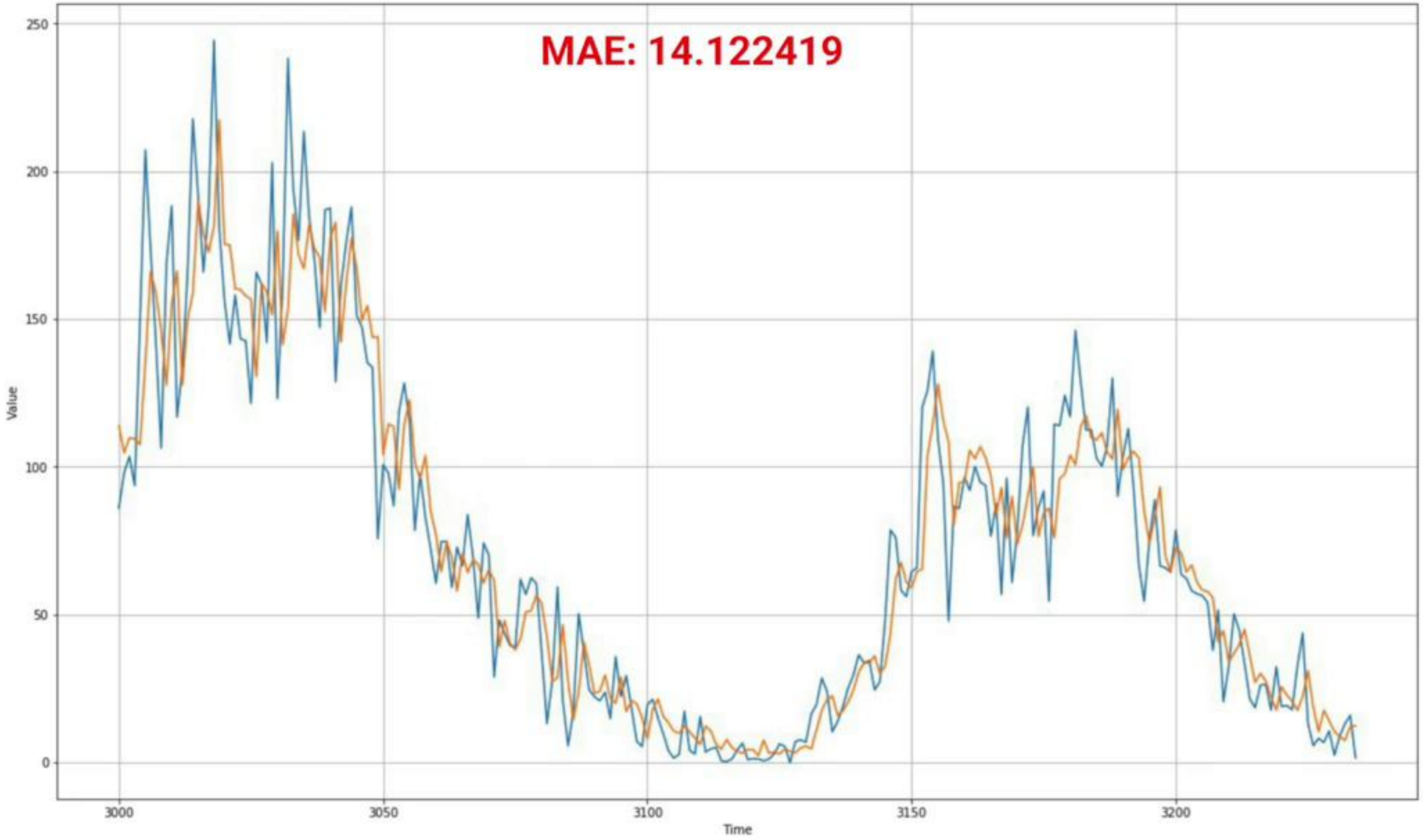
```
split_time = 1000
time_train = time[:split_time]
x_train = series[:split_time]
time_valid = time[split_time:]
x_valid = series[split_time:]

window_size = 30
batch_size = 32
shuffle_buffer_size = 1000
```


MAE: 15.1447315
250
200
150
100
50
0
Value
3000
3050
3100
3150
3200
Time

```
dataset = windowed_dataset(x_train, window_size, batch_size, shuffle_buffer_size)

model = tf.keras.models.Sequential([
    tf.keras.layers.Dense(10, input_shape=[window_size], activation="relu"),
    tf.keras.layers.Dense(10, activation="relu"),
    tf.keras.layers.Dense(1)
])

model.compile(loss="mse", optimizer=tf.keras.optimizers.SGD(lr=1e-6, momentum=0.9))
model.fit(dataset,epochs=100,verbose=0)
```


MAE: 14.348902
250
200
150
100
50
0
Value
3000
3050
3100
3150
3200
Time

```
dataset = windowed_dataset(x_train, window_size, batch_size, shuffle_buffer_size)

model = tf.keras.models.Sequential([
    tf.keras.layers.Dense(10, input_shape=[window_size], activation="relu"),
    tf.keras.layers.Dense(10, activation="relu"),
    tf.keras.layers.Dense(1)
])

model.compile(loss="mse", optimizer=tf.keras.optimizers.SGD(lr=1e-6, momentum=0.9))
model.fit(dataset,epochs=100,verbose=0)
```


MAE: 14.122419
250
200
150
100
50
0
Value
3000
3050
3100
3150
3200
Time

```
model.predict(series[3205:3235][np.newaxis])
```

https://www.sws.bom.gov.au/Solar/1/6

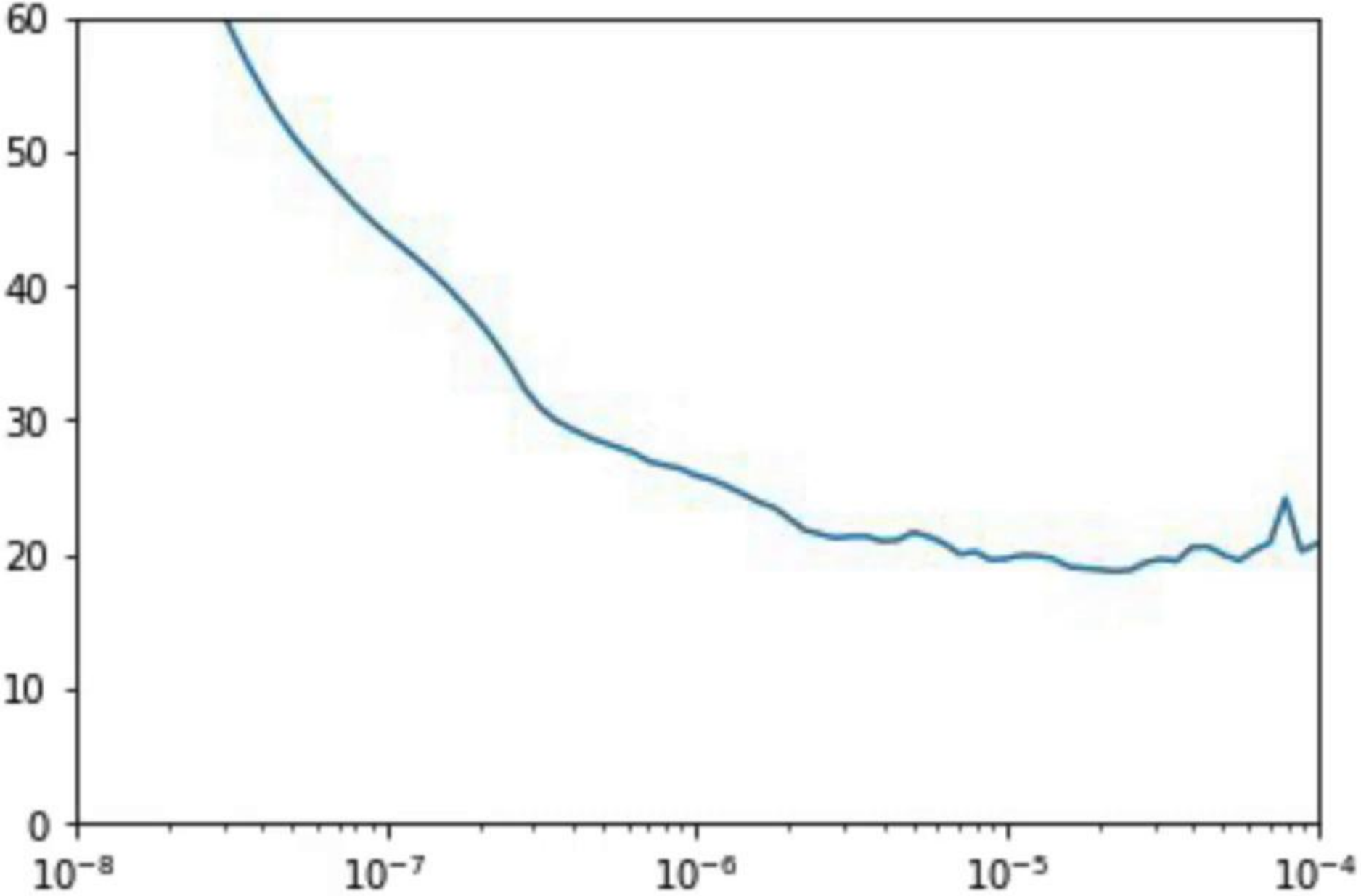
(last updated 01 Jun 2019 09:42 UT)

OBSERVED MONTHLY SUNSPOT NUMBERS

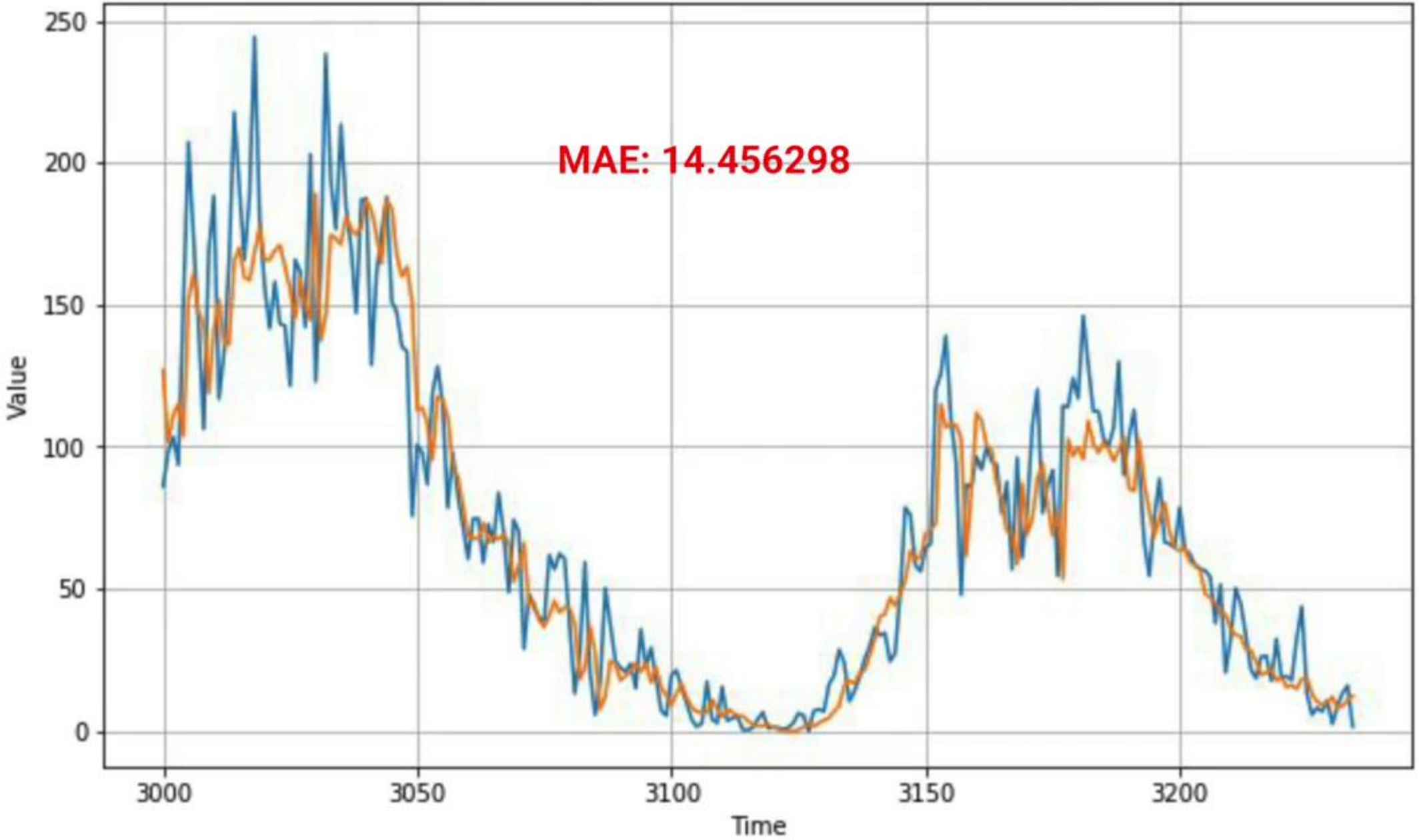
2001	142.6	121.5	165.8	161.7	142.1	202.9	123.0	161.5	238.2	194.1	176.6	213.4
2002	184.6	170.2	147.1	186.9	187.5	128.8	161.0	175.6	187.9	151.2	147.2	135.3
2003	133.5	75.7	100.7	97.9	86.8	118.7	128.3	115.4	78.5	97.8	82.9	72.2
2004	60.6	74.6	74.8	59.2	72.8	66.5	83.8	69.7	48.8	74.2	70.1	28.9
2005	48.1	43.5	39.6	38.7	61.9	56.8	62.4	60.5	37.2	13.2	27.5	59.3
2006	20.9	5.7	17.3	50.3	37.2	24.5	22.2	20.8	23.7	14.9	35.7	22.3
2007	29.3	18.4	7.2	5.4	19.5	21.3	15.1	9.8	4.0	1.5	2.8	17.3
2008	4.1	2.9	15.5	3.6	4.6	5.2	0.6	0.3	1.2	4.2	6.6	1.0
2009	1.3	1.2	0.6	1.2	2.9	6.3	5.5	0.0	7.1	7.7	6.9	16.3
2010	19.5	28.5	24.0	10.4	13.9	18.8	25.2	29.6	36.4	33.6	34.4	24.5
2011	27.3	48.3	78.6	76.1	58.2	56.1	64.5	65.8	120.1	125.7	139.1	109.3
2012	94.4	47.8	86.6	85.9	96.5	92.0	100.1	94.8	93.7	76.5	87.6	56.8
2013	96.1	60.9	78.3	107.3	120.2	76.7	86.2	91.8	54.5	114.4	113.9	124.2
2014	117.0	146.1	128.7	112.5	112.5	102.9	100.2	106.9	130.0	90.0	103.6	112.9
2015	93.0	66.7	54.5	75.3	88.8	66.5	65.8	64.4	78.6	63.6	62.2	58.0
2016	57.0	56.4	54.1	37.9	51.5	20.5	32.4	50.2	44.6	33.4	21.4	18.5
2017	26.1	26.4	17.7	32.3	18.9	19.2	17.8	32.6	43.7	13.2	5.7	8.2
2018	6.8	10.7	2.5	8.9	13.1	15.6	1.6	8.7	3.3	4.9	4.9	3.1
2019	7.8	0.8	9.5	9.1	10.1							

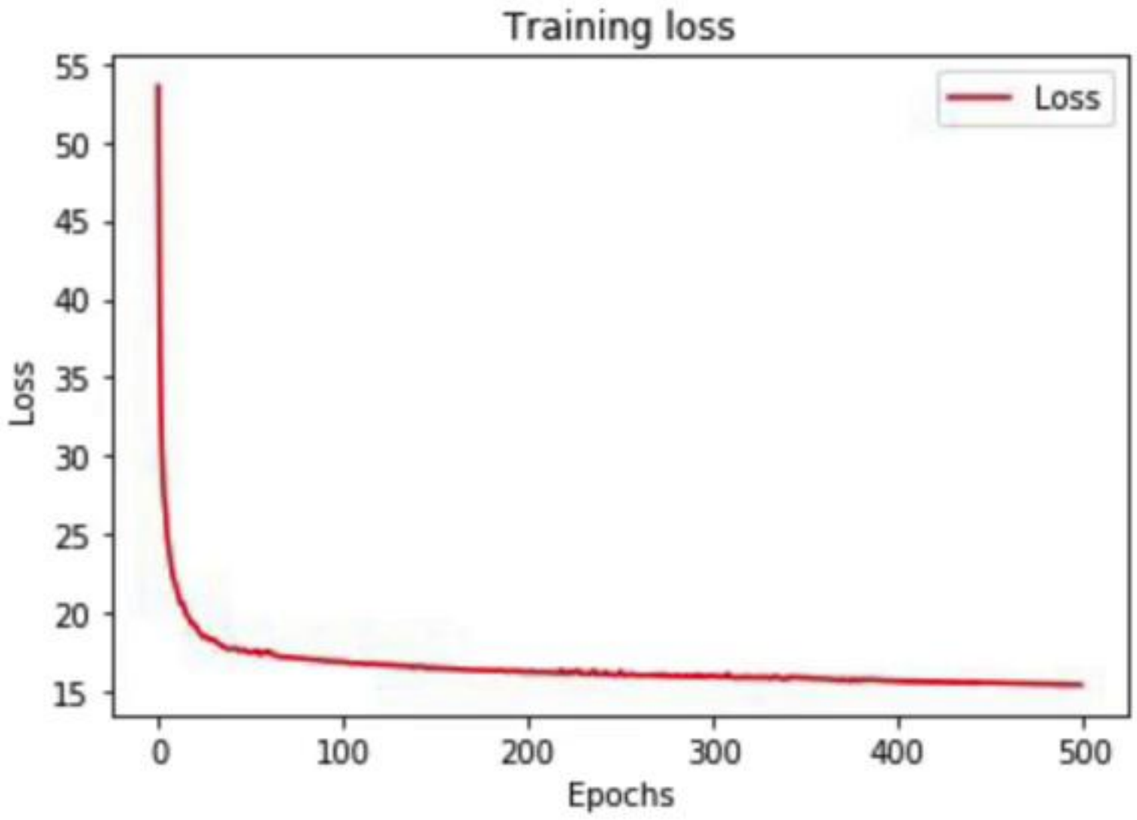
```python
split_time = 3000
window_size = 60

model = tf.keras.models.Sequential([
    tf.keras.layers.Dense(20, input_shape=[window_size], activation="relu"),
    tf.keras.layers.Dense(10, activation="relu"),
    tf.keras.layers.Dense(1)
])

model.compile(loss="mse", optimizer=tf.keras.optimizers.SGD(lr=1e-7, momentum=0.9))
```

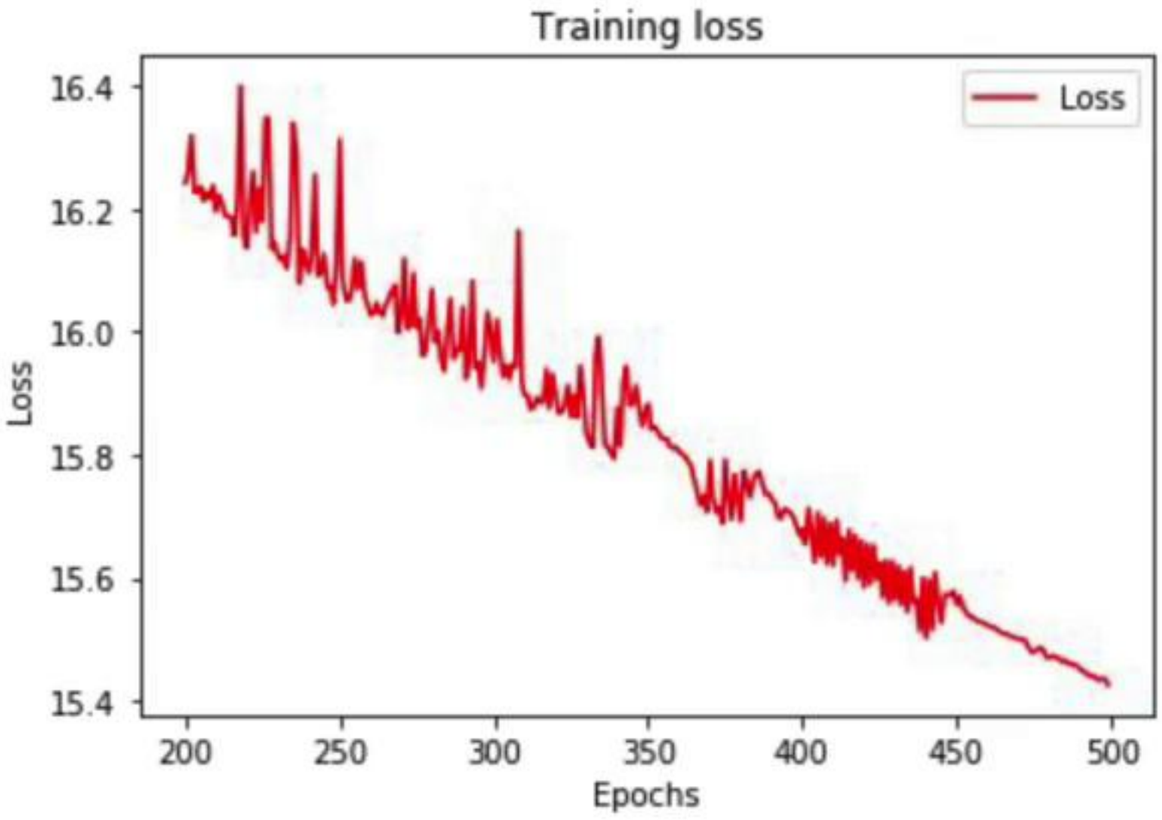
```python
window_size = 60
batch_size = 64
train_set = windowed_dataset(x_train, window_size, batch_size, shuffle_buffer_size)

model = tf.keras.models.Sequential([
  tf.keras.layers.Conv1D(filters=32, kernel_size=5, strides=1, padding="causal", activation="relu",
                      input_shape=[None, 1]),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.Dense(30, activation="relu"),
  tf.keras.layers.Dense(10, activation="relu"),
  tf.keras.layers.Dense(1),
  tf.keras.layers.Lambda(lambda x: x * 400)
])

lr_schedule = tf.keras.callbacks.LearningRateScheduler(lambda epoch: 1e-8 * 10**(epoch / 20))
optimizer = tf.keras.optimizers.SGD(lr=1e-8, momentum=0.9)
model.compile(loss=tf.keras.losses.Huber(), optimizer=optimizer, metrics=["mae"])
history = model.fit(train_set, epochs=100, callbacks=[lr_schedule])
```


60
50
40
30
20
10
0
10^{-8}
10^{-7}
10^{-6}
10^{-5}
10^{-4}

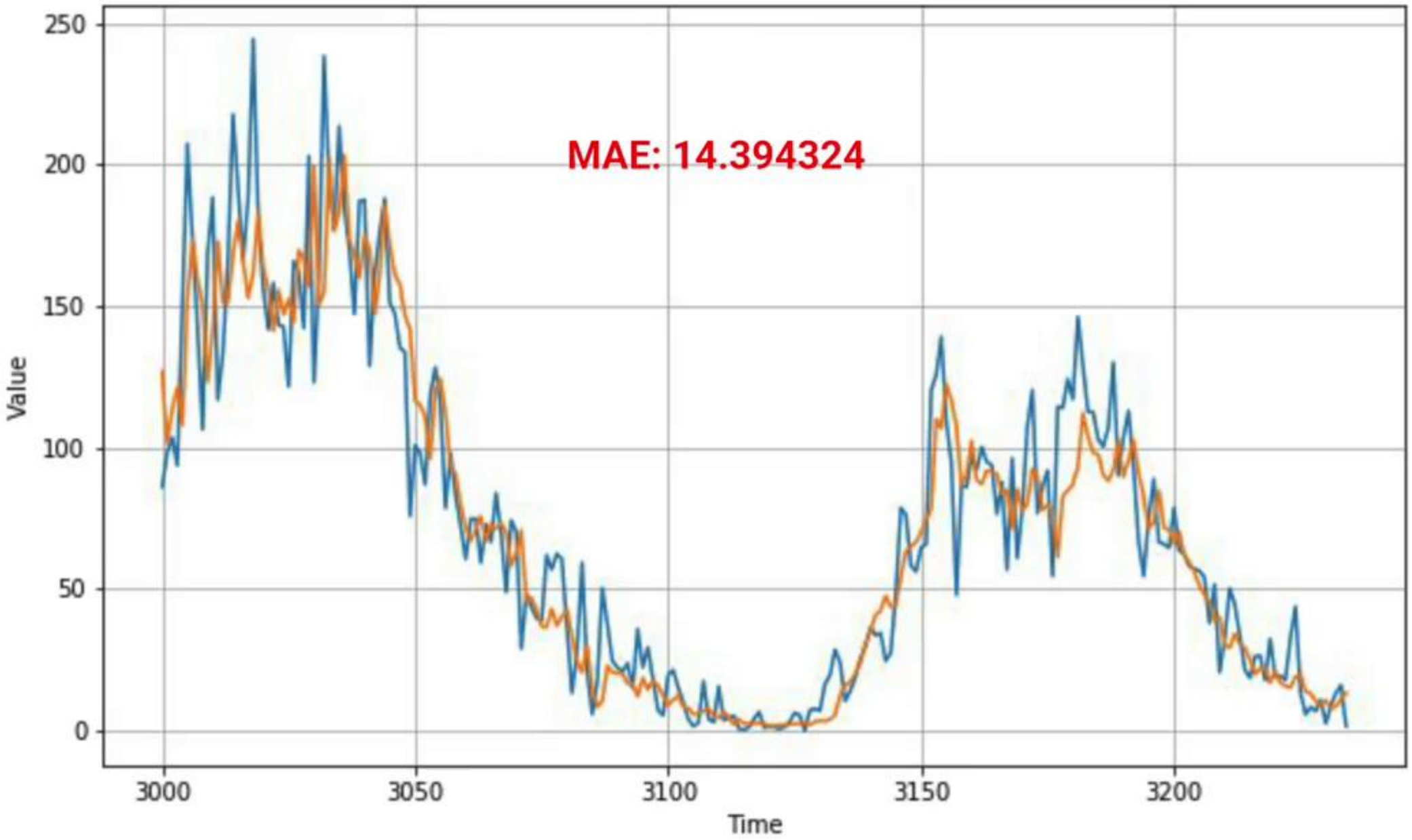
```
train_set = windowed_dataset(x_train, window_size, batch_size, shuffle_buffer_size)
model = tf.keras.models.Sequential([
  tf.keras.layers.Conv1D(filters=32, kernel_size=5,
                         strides=1, padding="causal",
                         activation="relu",
                         input_shape=[None, 1]),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.Dense(30, activation="relu"),
  tf.keras.layers.Dense(10, activation="relu"),
  tf.keras.layers.Dense(1),
  tf.keras.layers.Dense(1),
  tf.keras.layers.Lambda(lambda x: x * 400)
])


optimizer = tf.keras.optimizers.SGD(lr=1e-5, momentum=0.9)
model.compile(loss=tf.keras.losses.Huber(),
              optimizer=optimizer,
              metrics=["mae"])
history = model.fit(train_set,epochs=500)
```

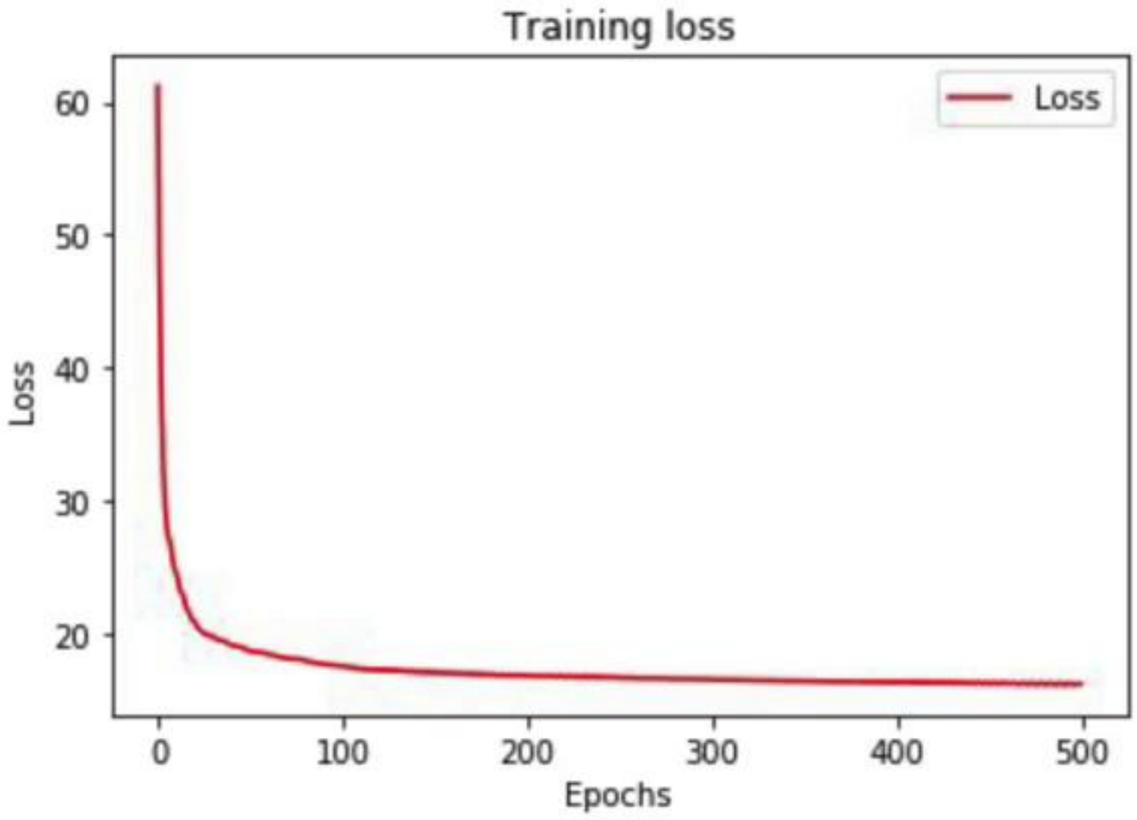

MAE: 14.456298
250
200
150
100
50
0
Value
3000
3050
3100
3150
3200
Time

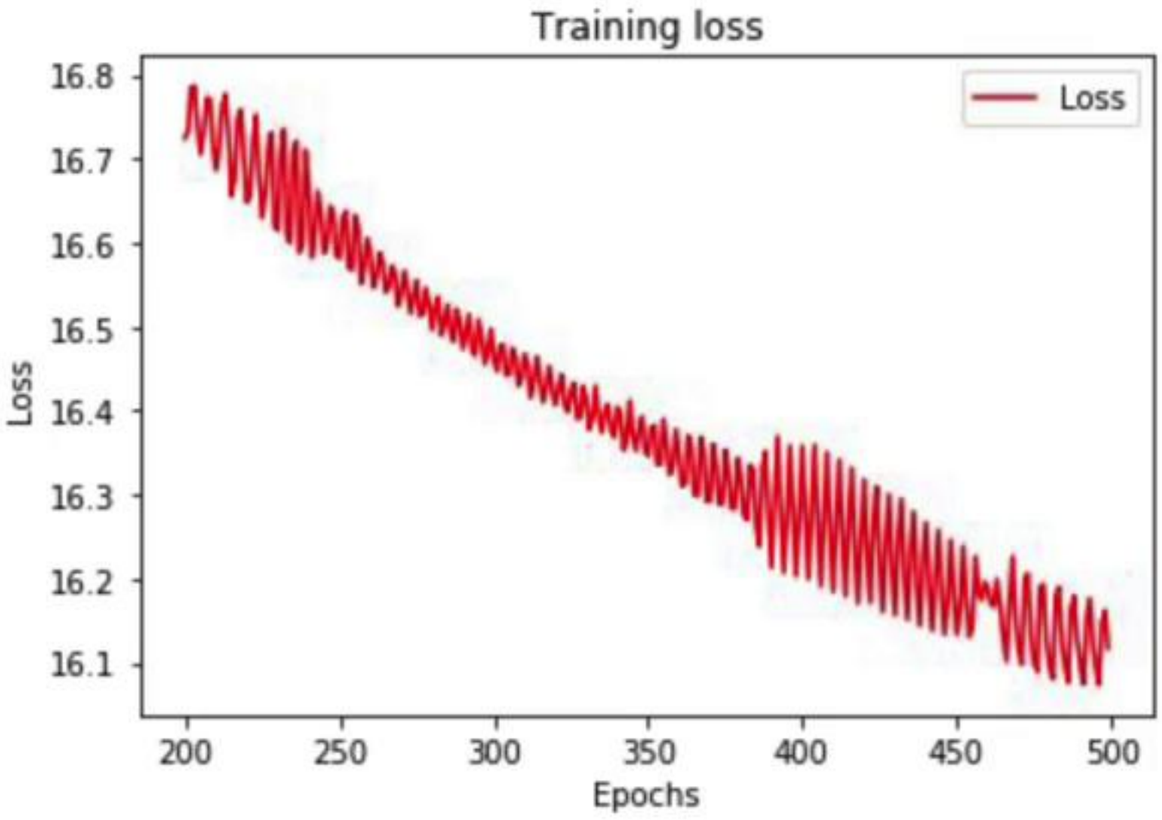

Training loss
Loss
55
50
45
40
35
30
25
20
15
0
100
200
300
400
500
Epochs
Loss


Training loss
Loss
16.4
16.2
16.0
15.8
15.6
15.4
200
250
300
350
400
450
500
Epochs
Loss

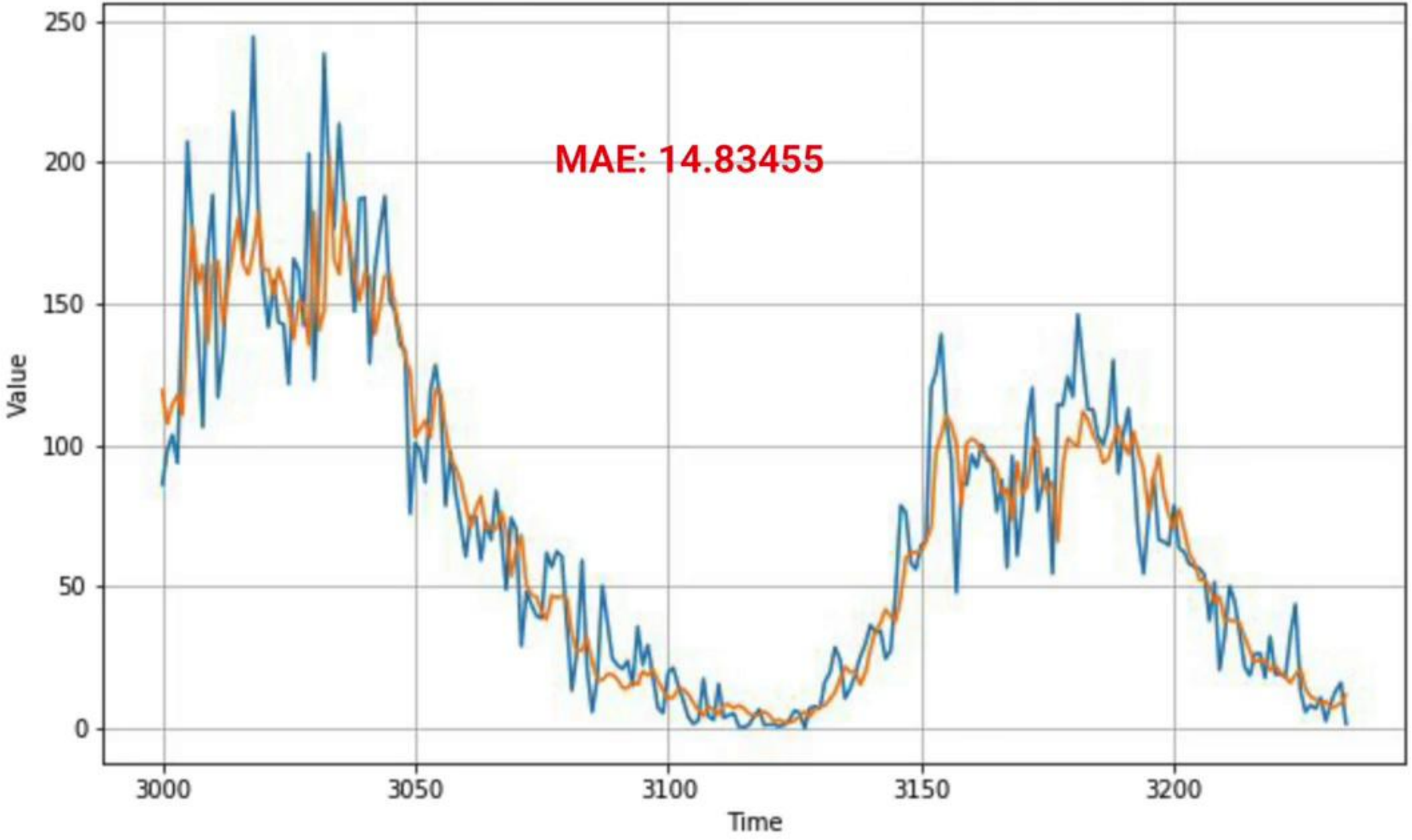
```
train_set = windowed_dataset(x_train, window_size, batch_size=256, shuffle_buffer_size)
model = tf.keras.models.Sequential([
  tf.keras.layers.Conv1D(filters=32, kernel_size=5,
                         strides=1, padding="causal",
                         activation="relu",
                         input_shape=[None, 1]),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.LSTM(32, return_sequences=True),
  tf.keras.layers.Dense(30, activation="relu"),
  tf.keras.layers.Dense(10, activation="relu"),
  tf.keras.layers.Dense(1),
  tf.keras.layers.Dense(1),
  tf.keras.layers.Lambda(lambda x: x * 400)
])


optimizer = tf.keras.optimizers.SGD(lr=1e-5, momentum=0.9)
model.compile(loss=tf.keras.losses.Huber(),
              optimizer=optimizer,
              metrics=["mae"])
history = model.fit(train_set,epochs=500)
```

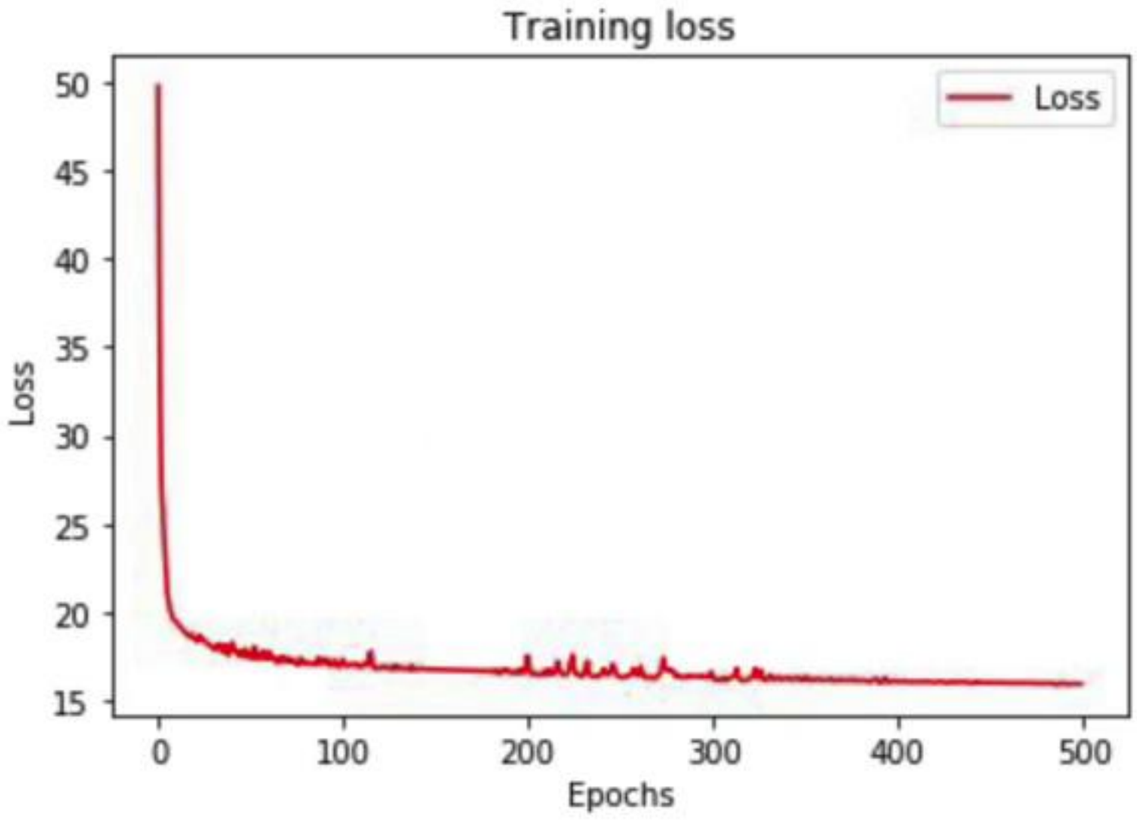

MAE: 14.394324
250
200
150
100
50
0
Value
3000
3050
3100
3150
3200
Time

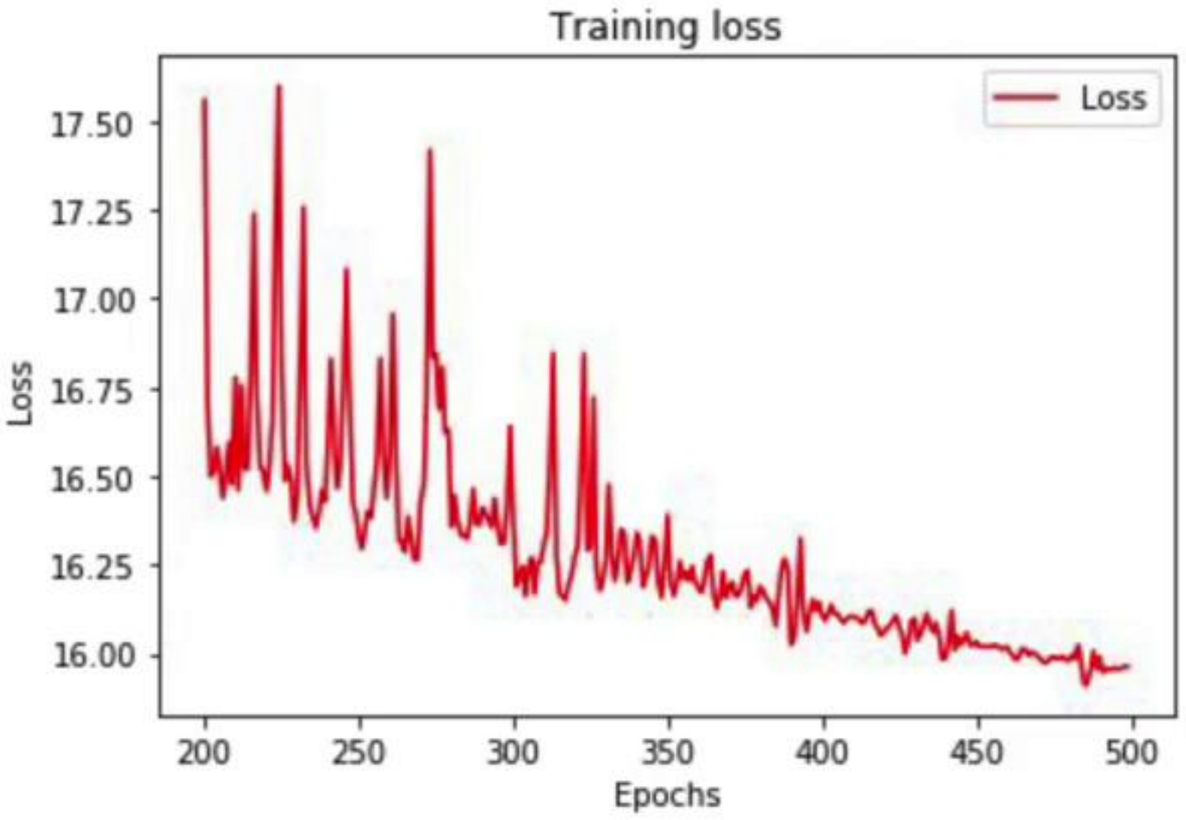

Training loss
Loss
60
50
40
30
20
Loss
0
100
200
300
400
500
Epochs

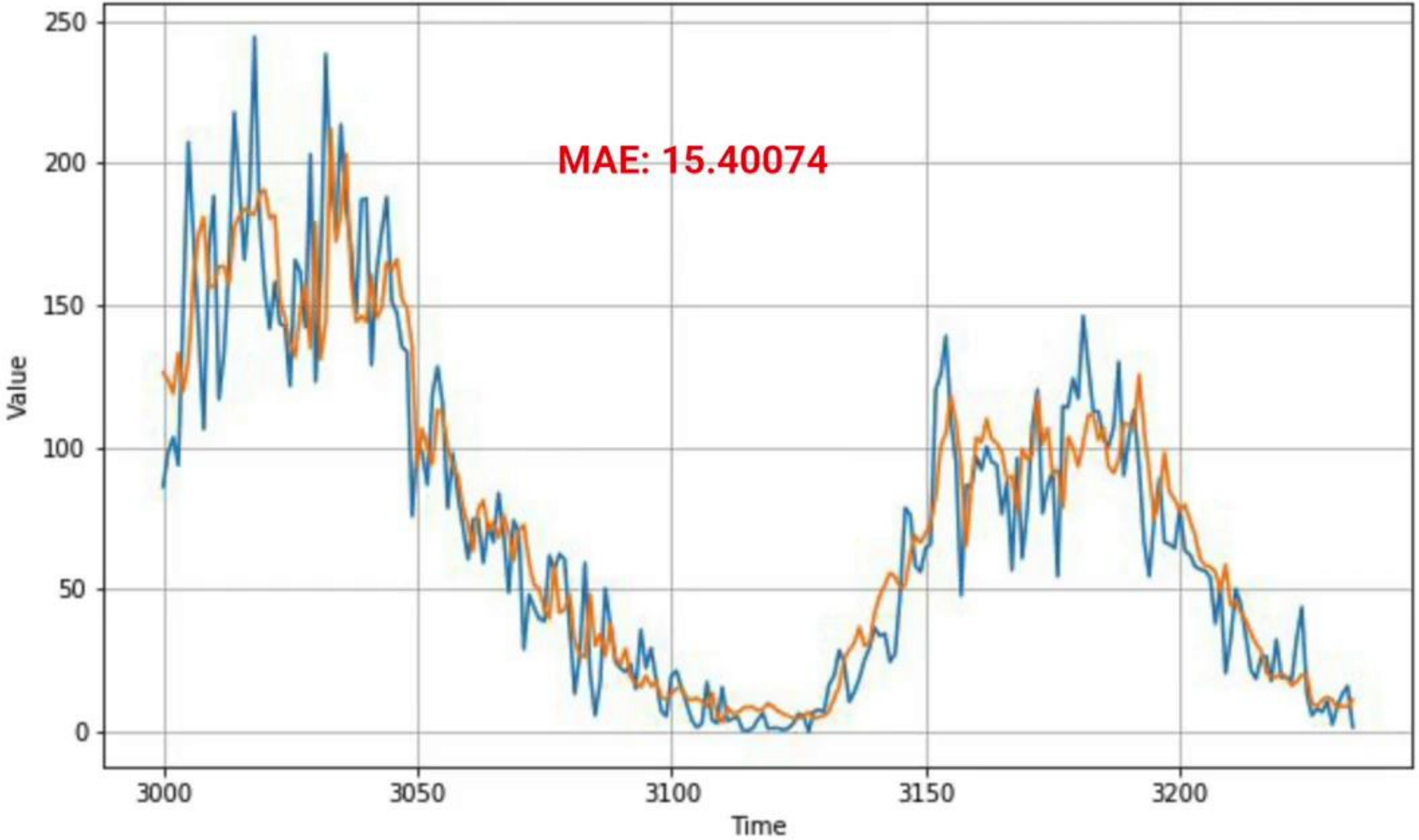

Training loss
Loss
16.8
16.7
16.6
16.5
16.4
16.3
16.2
16.1
Loss
200
250
300
350
400
450
500
Epochs

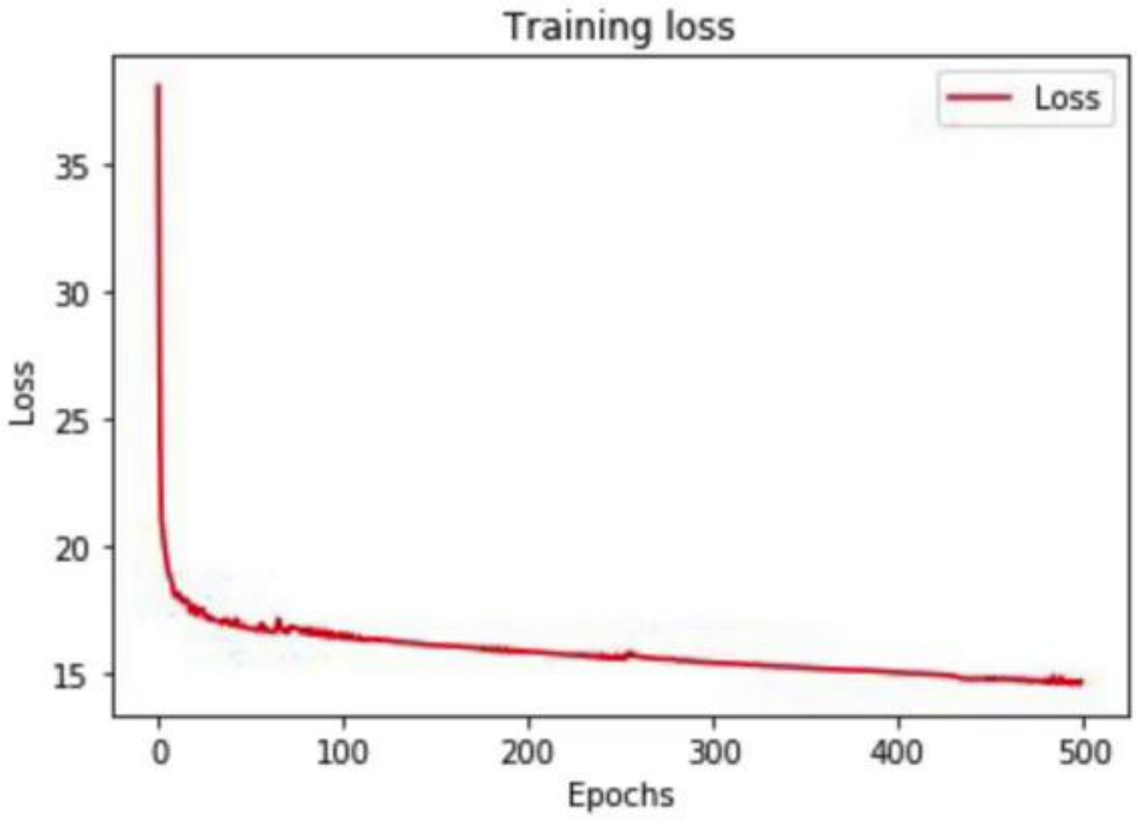
```
train_set = windowed_dataset(x_train, window_size=60, batch_size=250, shuffle_buffer_size)
model = tf.keras.models.Sequential([
  tf.keras.layers.Conv1D(filters=60, kernel_size=5,
                      strides=1, padding="causal",
                      activation="relu",
                      input_shape=[None, 1]),
  tf.keras.layers.LSTM(60, return_sequences=True),
  tf.keras.layers.LSTM(60, return_sequences=True),
  tf.keras.layers.Dense(30, activation="relu"),
  tf.keras.layers.Dense(10, activation="relu"),
  tf.keras.layers.Dense(1),
  tf.keras.layers.Lambda(lambda x: x * 400)
])


optimizer = tf.keras.optimizers.SGD(lr=1e-5, momentum=0.9)
model.compile(loss=tf.keras.losses.Huber(),
              optimizer=optimizer,
              metrics=["mae"])
history = model.fit(train_set,epochs=500)
```

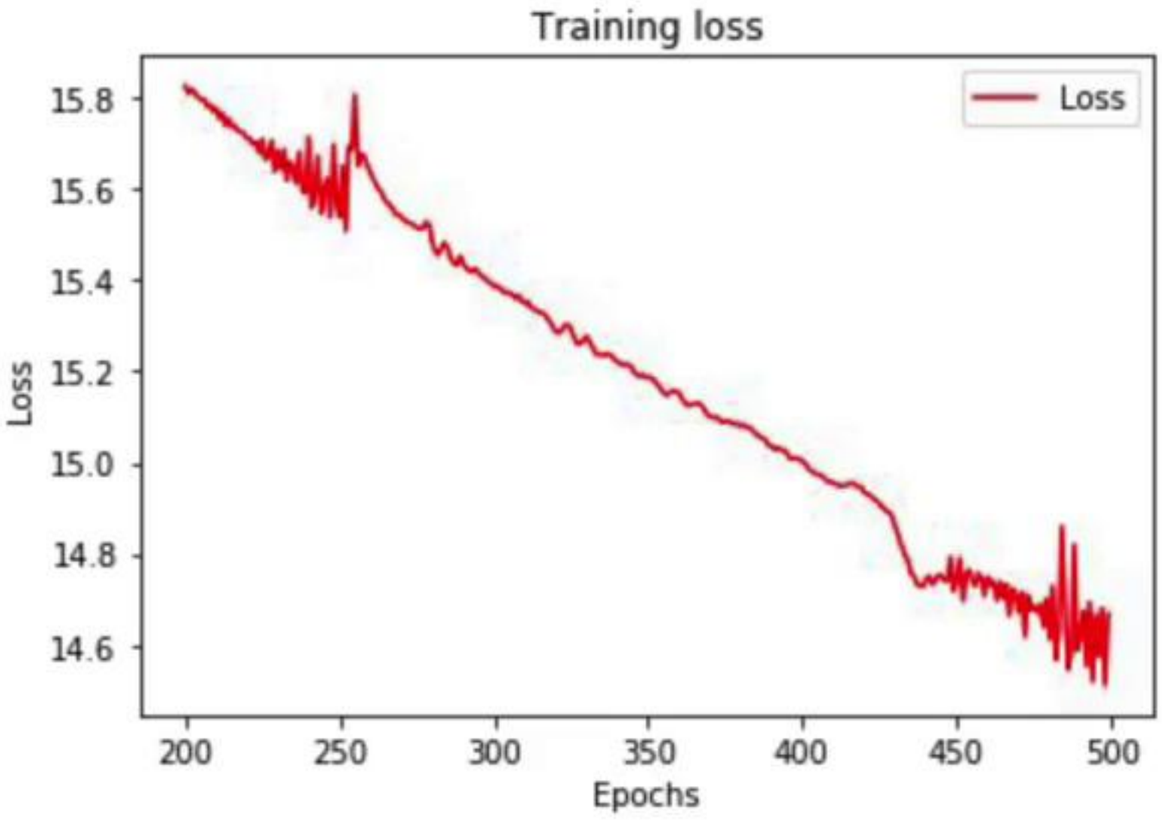

MAE: 14.83455
250
200
150
100
50
0
Value
3000
3050
3100
3150
3200
Time


Training loss
Loss
50
45
40
35
30
25
20
15
0
100
200
300
400
500
Epochs


Training loss
Loss
17.50
17.25
17.00
16.75
16.50
16.25
16.00
200
250
300
350
400
450
500
Epochs


MAE: 15.40074
250
200
150
100
50
0
Value
3000
3050
3100
3150
3200
Time


Training loss
Loss
35
30
25
20
15
Loss
0
100
200
300
400
500
Epochs


Training loss
Loss
15.8
15.6
15.4
15.2
15.0
14.8
14.6
Loss
200
250
300
350
400
450
500
Epochs